

Applications of AI in martial arts: A survey

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Abstract

Artificial intelligence (AI) has become an interdisciplinary subject that integrates digital image processing, machine learning, and computer science, etc. A lot of work in martial arts benefits from AI technology. In this paper, we highlight and summarize the **applications of AI in martial arts** and indicate the main progress of related studies. Through the investigation of martial arts styles, tasks, data acquisition methods, and algorithm improvements, a paradigm that represents a possible technological process for an intelligent martial arts training system was summarized. These applications bring more scientific training and more effective technical analysis to martial arts. Furthermore, this paper delves into specific AI applications in areas such as action recognition, pose estimation, action evaluation, elite athlete support, sports betting, health care, animation generation, and competition field segmentation. By exploring these areas, the research prospects of AI applications in martial arts are comprehensively discussed. This paper inspires further research on martial arts by using AI, which includes helping researchers expand their research ideas and research methods, identify frontier issues, and innovate based on existing research.

Keywords

Martial arts, artificial intelligence, sensor, computer vision, human-computer interaction, pose estimation, machine learning, ai, sports

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Introduction

Martial arts are any form and skill of fighting and are systems of fighting developed for a variety of reasons, including self-defense.¹ Kung fu is a similar concept to martial arts, and when referring to kung fu, the tendency is to emphasize the intrinsic nature of martial arts.² Martial arts can be divided into armed and unarmed categories.³ For unarmed martial arts, some specialize in striking techniques (e.g. boxing, kickboxing, taekwondo), some specialize in grappling (e.g. wrestling, sumo, judo), some specialize in joint locking techniques (e.g. aikido, Qinna), or a combination of several specializations (e.g. mixed martial arts, Sanda, Muay Thai).^{4,5} Martial arts also include skills in the use of cold weapons, which can be categorized as long weapons, short weapons, soft weapons, projectile weapons, etc.⁶ Self-defense is the most primitive and basic function of martial arts.⁷ Any form of martial arts is combative in nature, and can make the opponent lose the ability to resist and play the role of self-defense. As society progresses, participation in martial arts for health is becoming more and more important,

and studies have shown that practicing martial arts can help reduce mental stress, improve flexibility and balance, posture control, and lower extremity strength, prevent the decline of the musculoskeletal system of the elderly and reduce the possibility of falling.^{8–11} For martial artists to gain self-defense and health functions, they must practice or use martial arts based on proper and rational practice.

In recent years, Artificial intelligence (AI) has developed rapidly in computer vision, signal processing and other areas. The application of various machine learning algorithms in sports is becoming more and more widespread and in-depth. Martial arts have diversity

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and complexity, encompassing both non-interactive (e.g. some solo routines) and interactive (partner routines, combat). Applying AI techniques to martial arts is a fascinating task. AI algorithms bring enormous benefits to martial arts, including more scientific training and more effective technical and tactical analysis. Although applying research on related aspects has started early, it has not been comprehensively summarized. Yuqing et al.¹² and Pang et al.¹³ discussed and summarized the application of **virtual reality** and computer vision in **martial arts**; Ke¹⁴ and Zhang¹⁵ discussed the application of AI in martial arts events. However, a comprehensive overview of the broader applications of AI in martial arts is currently needed to help people understand this field more rapidly. This paper is devoted to providing a summary and review of these applications to facilitate future research efforts. Specifically, we:

- (1) Summarize the general paradigm of research on these applications, which embodies the possible use of technical processes as well as optimization ideas for intelligent martial arts training systems.
- (2) **Collate open-source datasets** related to martial arts and analyze current research hotspots and content.
- (3) Investigate the AI application of martial arts in not only action recognition, pose estimation, and action evaluation, but also serving elite athletes, sports betting, health care, animation generation, and competition field segmentation.
- (4) Count and analyze publications from the perspective of volumes of publication, martial arts styles and algorithms, and to apply trends in AI to martial arts.

To sum up, this paper takes a comprehensive look at the applications of AI within the domain of martial arts. Through a thorough search methodology, we have compiled and analyzed diverse literature, covering aspects such as action recognition, pose estimation, **open-access datasets**, and various other applications. As we delve into the subsequent sections of this paper, our objective is to unveil the innovative ways in which AI is reshaping the landscape of martial arts. By shedding light on the current state of research and exploring potential future directions, we aim to provide valuable insights for researchers, practitioners, and enthusiasts alike.

The rest of the article is organized as follows: Section II describes the research method, mainly referring to searching the literature. Section III delves into the records obtained through the literature search, where the statistics of the number of documents and an analysis of types and styles of martial arts within the literature are thoroughly explored. **Section IV provides a valuable resource for researchers and practitioners; this section highlights the importance of open-access datasets in advancing AI applications within martial arts.** Section V explores various methods for action recognition in martial arts. Section V encompasses image/

visual feature-based methods, human skeleton-based methods, and the emerging field of action recognition using wearable devices. **In Section VI, optimization techniques in pose estimation are covered.** Section VII addresses techniques related to action evaluation, including template methods, classification, and regression approaches. Beyond action recognition and pose estimation, **Section VIII** explores diverse applications of AI in the martial arts domain, showcasing its versatility. In Section IX, we engage in a comprehensive discussion, this section delves into suitable scenarios for AI algorithms, rethinking the status of action recognition in AI-martial arts, exploring the concept of an intelligent martial arts system, and discussing the potential focus of future research. Finally, Section X summarizes our findings, this section offers key takeaways from the study and provides insights into the current state of AI in martial arts. These are graphically demonstrated in Figure 1.

Search methodology

Search strategy

Two digital databases (Web of Science Core Collection, EI Compex) were searched for publications dated from January 1, 2013, to December 31, 2023. In the category of artificial intelligence, the search terms are: artificial intelligence, deep learning, machine learning, recognition, and pose estimation; in the category of martial arts, the search keywords are: martial arts, wushu, Tai Chi, taiji, taekwondo, karate, etc., as detailed in the Supplemental Materials. The keywords from the two aspects are combined with each other. As an example, search in the Web of Science Core Collection database: ALL=((“martial arts”) AND (“Artificial intelligence” OR “Deep Learning” OR “Machine Learning” OR “Computer Vision” OR “Recognition” OR “Pose estimation”)). Dozens of search queries were applied to include as many martial art types as possible. Records of conference articles, journal articles, preprinted articles, and theses are selected. After deleting duplicates, the researchers filtered the identified materials according to the titles and abstracts.

Inclusion and exclusion criteria

The exclusion criteria are: (1) the language of the article is not English or Chinese; (2) the article does not include machine learning or deep learning algorithms; (3) the source of the data in the article is not clear; (4) the proposed methodology does not compare with other methodologies.

Information abstract and analysis

After screening, the information extracted from the selected publications includes: (1) author(s) and

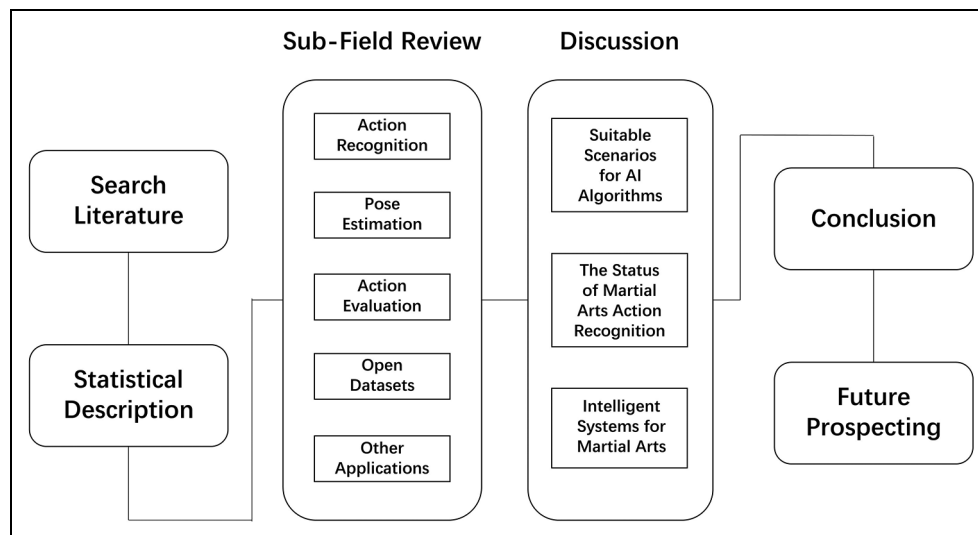


Figure 1. The flow of how the survey is organized.

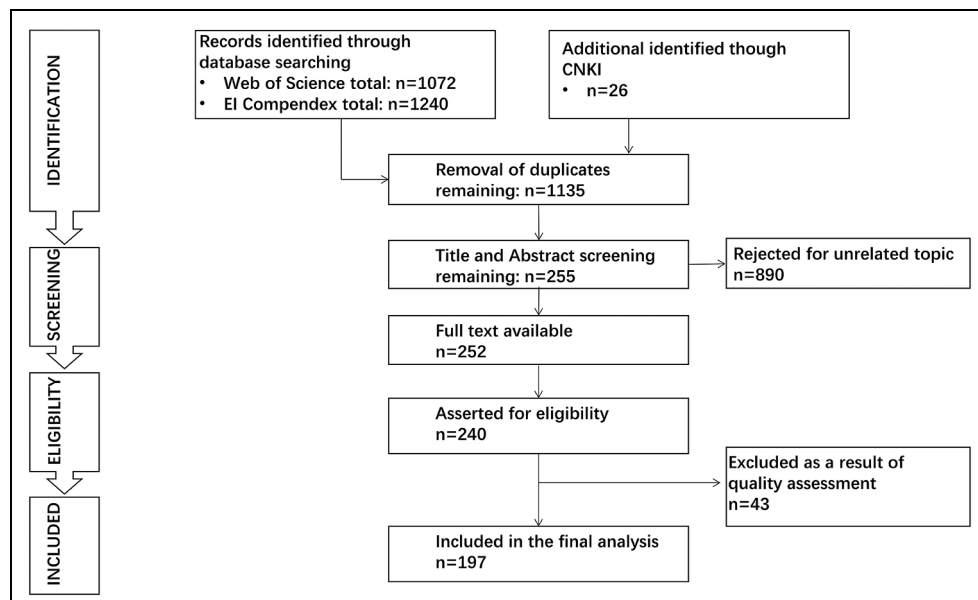


Figure 2. Literature screening process and results.

publication date, (2) main algorithm(s) and optimization ideas, (3) data collection equipment, (4) samples and subjects, and (5) the types of martial arts.

Searched records

Searched records and screening

The searched records of the Web of Science database amounted to 1071. In the EI Compindex database, 1240 records were identified. Another 26 records supplemented from China national knowledge internet (CNKI) were also included. After deleting duplicates, 1135 records were retained. We screened 255 records according to their titles and abstracts. Of the remaining

255 records, 252 records are available as full-text. During the eligibility check, 43 records were excluded according to the exclusion criteria. Therefore, 197 studies were included in the manuscript (Figure 2).

Quantitative characteristics of the literature

Statistics of the documents. As shown in Figure 3, from 2013 to 2023, the number of AI in martial arts paper publications shows a trend of steady growth followed by a significant increase. Between 2013 and 2016, the number of paper publications remained stable. However, starting in 2017, the number of paper publications began to grow significantly, reaching 22 in 2019

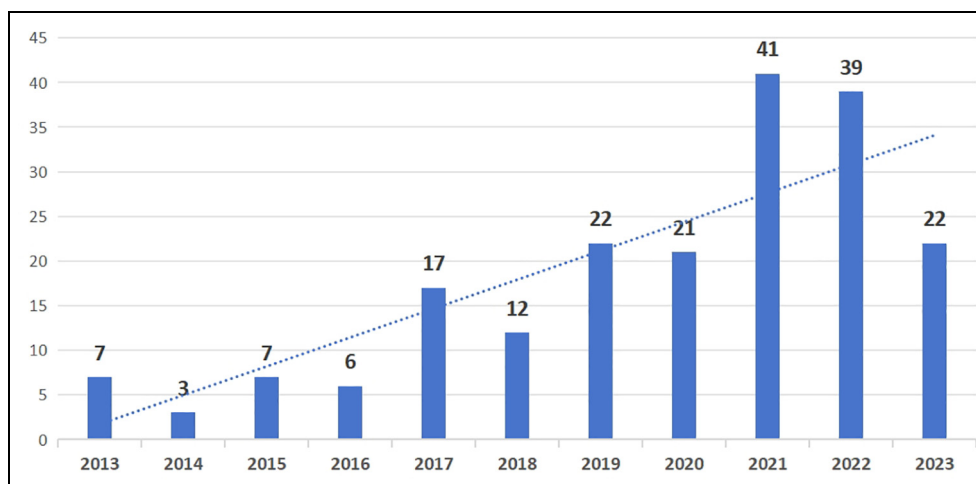


Figure 3. Number of publications from 2013 to 2023.

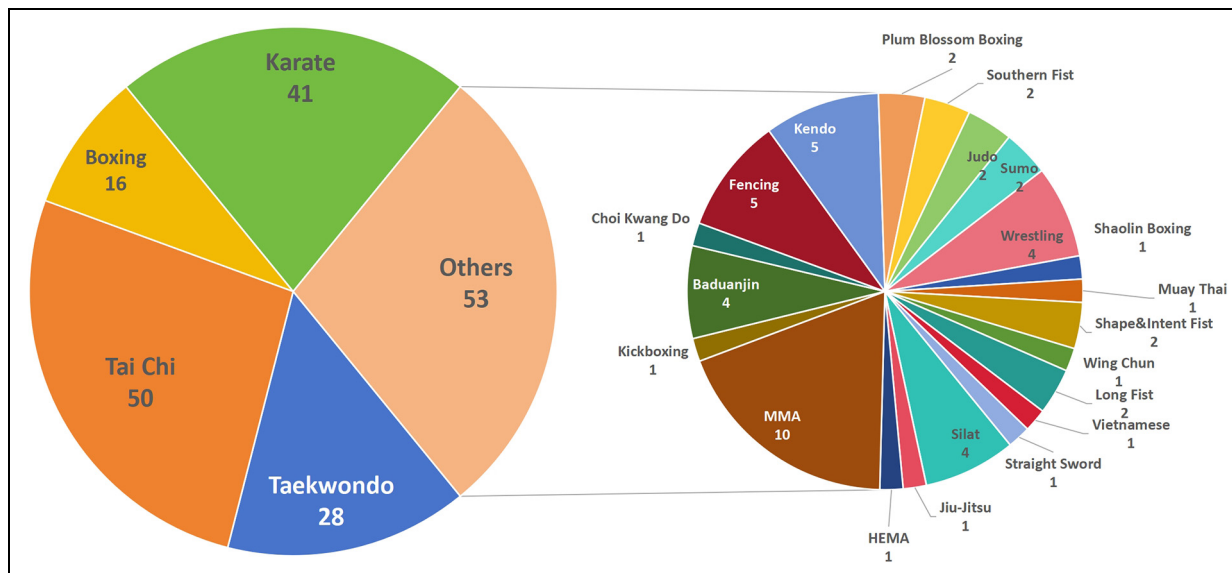


Figure 4. The number of applications of AI in various martial arts.

and soaring to 41 in 2021. By 2022, the number of papers issued remained high at 39, albeit falling back. This trend shows that as AI technology continues to evolve, researchers are showing a strong interest in how this technology can be applied to martial arts. At the same time, it also reflects that academic research in this field is gradually deepening.

In the future, with the further development of AI technology and the continued exploration of the martial arts community, we believe that the number of papers issued in this field will continue to grow. Meanwhile, it is expected to see more high-quality research results, which will provide strong support for promoting the application of AI in martial arts.

Types and styles of martial arts. As shown in Figure 4, these studies cover a variety of martial arts, but most

of them focus on martial arts like Tai Chi, Taekwondo, Karate, and Boxing, which are widespread, while there are fewer studies on other relatively niche martial arts.

Open access dataset

Data is one of the key drivers of modern AI. We investigate datasets about martial arts, and extract information such as year, type of martial art, data acquisition method, and task. Detailed descriptions of open access datasets can be found in Table 1.

Action recognition in martial arts

Human action recognition (HAR) takes up a large portion of the applications of artificial intelligence in martial arts. It categorizes the kinds of actions for people to

Table 1. An overview of open access dataset recognition.

Name	Year	Task	Data acquisition modality	Description	Types of martial arts
UMONS-TAICHI	2018 ¹⁶	Action recognition	11 Qualisys cameras (68 markers, 179 Hz); One Kinect V2 RGB-D sensor (tracking 25 key points, 30 Hz)	12 practitioners of different levels, eight techniques and five kinds of footwork, 2200 records in total	Tai Chi
MADS	2015 ¹⁷	Pose estimation	Marker-based MOCAP system capturing 19 key points, three RGB cameras, one Stereo depth camera	MADS is a high-quality 3D gesture dataset; eight actions in Tai Chi and six actions in Karate	Chen style Tai Chi, karate
HPMRSS	2016 ¹⁸	Action recognition	One Kinect sensor	10 athletes of different genders, ages, and heights perform seven different actions. Athlete age range: 24–26. Athlete height range: 160–176 cm	Taekwondo
TUHAD	2020 ¹⁹	Action recognition	One mono RGB-D sensor	10 experts performing eight taekwondo moves, 99,982 frames in total (RGB, depth, infrared)	Taekwondo
HQMC	2017 ^{20,21}	Action recognition, motion capture visualization website	Wireless Shadow2.4 MoCap system.	It is a full-body, 6-DOF tracking solution with 17 inertial sensors and two pressure sensors. 100 Hz. Tracking frequency. Over 10,000 action samples	Karate
PressNet	2020 ²²	Predicts stability from 3D coordinates. Can also be used for pose estimation	12 marker-based MoCaps recording 25 key points, two RGB cameras, foot pressure insoles	Synchronized data of 1.36 million postures + plantar pressure from 10 subjects, 5 min duration for each	Tai Chi
UFC Stas	2018, ²³ 2019 ²⁴	Prediction of the winner of the fighting competition	Collected from the UFC stats website	Statistics on the red and the blue sides of the past Ultimate Fighting Championship (UFC)	Mixed martial arts
AMQ	2019 ²⁵	Action recognition and evaluation	25 body joints 3D positions with 250 Hz by Qualisys motion capture system	Full-body movements of seven athletes of different levels performing katas. 28 sequences in total. Katas scores were ranked by 14 experts according to the performances	Karate katas
Boxpunch	2019 ²⁶	Action recognition	Three-axis oriented accelerometer	The 7606 punches thrown by the eight boxers contained both sides of the frontal punch, hooks, the upper-cut and no-action	Boxing
ImmerTai DAKT	2018 ²⁷ 2021 ²⁸	VR system, 3D rebuilding Action recognition and evaluation	Kinect sensor Accelerator, RGB video	Eight Tai Chi actions 279 Anomalous kicks, 285 normal kicks	Tai Chi Taekwondo
Thai-Chi-HD	2019 ²⁹	Video driven animation	Collected from YouTube	280 Tai Chi videos. 252 videos for training, 28 for testing	Tai Chi

(continued)

Table 1. (continued)

Name	Year	Task	Data acquisition modality	Description	Types of martial arts
Kyokushin karate	2021 ³⁰	Classification, prediction and synthesis of human movement	Vicon Motion System sampling at 250 Hz; 39 reflective markers	Training kick or punch in the air/ at a target/ against an opponent	Karate
Straight Punch	2021 ³¹	Action evaluation	Pressure sensor; accelerometer; gyroscope	30 high-level boxers perform straight punch.	Boxing
HIDCL Tai Chi	2022 ³²	Action segment	15 joints trajectory collect by IUM at 125 Hz	20 action sequences performed by one male subject and one female subject	Tai Chi
Jiu-Jitsu Positions	2022 ³³	Pose estimation, action recognition	Collected from YouTube	120,279 labeled images of two athletes sparring in 10 combat positions and 18 keypoints	Brazilian Jiu-Jitsu
MATDAT	2023 ³⁴	Highlight detection	Acquired by a camera of iPhone 6S	More than 90 K labeled images corresponding video sequences that represent some common scenarios in martial arts tricks	Non-specific

learn and understand. Based on the action recognition, we can further distinguish whether the actions are correct or wrong. Different from image recognition, action recognition is more challenging because it needs to capture the temporal contexts within body movements. The methods of martial arts action recognition can be divided into three main methods: (1) image/visual feature-based methods, (2) human skeleton-based methods, and (3) wearable sensor-based methods.

Image/visual feature-based methods

Sun et al.³⁵ took dense trajectories to represent motion information, combined with the Gaussian Mixture Model (GMM), to achieve the fine-grained classification of 58 types of Tai Chi movements. The basic idea of the Dense Trajectory³⁶ algorithm is to use an optical flow field to obtain some trajectories in the video sequence, and then extract histogram of gradient (HOG), histogram of flow (HOF), and motion boundary histograms (MBH), along the trajectory. The HOF, HOG, MBH, and trajectory features are then extracted along the trajectory, where HOF is calculated based on the greyscale map and the other ones are calculated based on the dense optical flow. Finally, the features are encoded by the Fisher vector (FV), and then the classifier is trained based on the encoding information.

Hoshino et al.³⁷ used the difference of adjacent frames to represent the movements as a feature map input to a convolutional neural network (CNN), and the classification model can be used for good recognition. This method was achieved for kendo movements targeting face, throat, and abdominal attacks captured by small cameras at low frame rates (30 fps), and the adjacent three-frame differential was found to outperform the adjacent two-frame differential due to capturing longer temporal information. Wu and Koike³⁸ proposed FuturePose with the keypoint lattice-optical flow (KL-OF). This action recognition module is a dual-stream long-short term memory network (LSTM) that takes KL-OF sparse optical flow as input. and obtains image coordinates of key points of the human skeleton through the VNect pose estimator. After 3D reconstruction, students can learn the coach's boxing motions via VR glasses in real time.

Hörhan and Eidenberger³⁹ proposed a feature extraction approach to distinguish brawl videos from martial arts videos by including more shaking when filming as one of the features of brawl videos, extracting features based on scale-invariant feature transform matching (SIFT) and Gestalt interest points (GIP), and using SVM to distinguish brawls from martial arts (wrestling, sumo, boxing, kickboxing, and karate), achieving significantly better results than only classification results relying on color features (blood-stains) alone. Mishra and Srinivasa⁴⁰ used local temporal features (spatiotemporal interest points) to recognize competition videos of taekwondo and judo. The detected local features are essentially improved Harris detectors

in three dimensions, with time in the third dimension. And the performance of k-nearest neighbors (KNN) versus SVM as a classifier is also compared. Although SIFT has unparalleled advantages in invariant feature extraction from images, it is not perfect and still suffers from poor real-time performance, such as fewer feature points and the inability to extract feature points accurately for targets with smooth edges.

In contrast to manually crafted feature extractors, CNNs possess the inherent capacity to autonomously learn and extract features from images. This inherent capability substantially amplifies the model's aptitude for generalization. Wang⁴¹ designed a martial arts action recognition algorithm based on image recognition. By employing edge detection techniques, it successfully captured the contour edges of the human body, extracting the boundaries of each video frame and achieving their accumulation within a unified image. Utilizing this accumulated edge image, a grid-based direction gradient histogram was computed, thereby deriving the feature vector for action recognition. Subsequently, the enhanced dynamic time warping method was applied to calculate the similarity between various martial arts action feature vector sequences, resulting in a significantly superior recognition rate compared to the motion history image approach. On the basis of object detection, Sun et al.⁴² took the brightness, contrast, and color information of the target area as the input of a multi-layer perceptron to identify the names of key actions of the tai chi competition video. CNNs can extract the features automatically, thus avoiding explicit feature extraction. Poulou et al.⁴³ trained an image recognition model for taekwondo blocks and kicks by VGG-13, although this work did not incorporate temporal information. Lee and Jung¹⁹ used a mono RGB-D camera to record the RGB, depth, and infrared information of taekwondo actions at the same time, then took them as the input of CNN to identify keyframes. In Kong et al.,⁴⁴ a single frame motion recognition model was established, and SVM was used to capture long-term information without extracting motion features from video sequences.

However, various image/visual feature-based methods are less robust to background changes and are gradually inferior to skeleton feature-based methods with the emergence of high-quality motion capture technologies.

Human skeleton-based methods

Skeleton-based Action Recognition aims to identify the action being performed from a series of temporally continuous human skeletal points. The human skeleton modality is naturally closer to human action and more compact than RGB frames or optical flow. Thus, the human skeleton modality is widely used in various action recognition tasks, and this modality tends to be lighter and more generalizable than algorithms based on other modalities. Compared to image features, joint

coordinates can better represent the kinematic features of the human body and have a stronger semantic meaning. Action recognition based on joint coordinates often uses RGB-D sensors, pose estimation, or wearable sensors to acquire joint coordinates and then import these kinematic features into supervised models, such as SVM, HMM, LSTM, MLP, and few-shot methods, such as gesture description language (GDL), dynamic time wrapping (DTW). However, due to the limitation of information contained in skeletal points, it is difficult for skeletal point-based algorithms to effectively recognize some actions that are closely related to objects or scenes.

With the popularity of stereo vision, the RGB-D sensor is involved in recognizing martial arts actions. In Zhang et al.,⁴⁵ the Kinect camera was able to extract the human skeletal key points and use principal component analysis (PCA) for dimensionality reduction and noise reduction. Thus, the 24-style Tai Chi movements were successfully recognized by naive Bayesian nearest neighbors. Salazar et al.⁴⁶ obtained the coordinates of human skeletal key points using the Kinect RGB-D camera, which represented the motion by main joint angles, and input the data to a SVM classifier to achieve image recognition of three Taekwondo actions: "chop," "bow step and down block" and "cross-step punch" (without modeling the temporal sequence). A sample of 1200 segments was captured with the Kinect V2 in Hachaj et al.,⁴⁷ including four defensive techniques (gedan-barai, jodan-uke, soto-uke, uchi-uke), three types of kicks (hiza-geri, mae-geri, and yoko-geri), and three standing stances (kiba-dachi, kokutsu-dachi, and zenkutsu-dachi). Several HMM classifiers with different numbers of hidden states as well as GMM classifiers were also used to empirically find the best hyperparameters for the proposed method in the dataset. Based on the key pose and latent dynamic conditional random field (LDCRF), de Souza Vicente et al.¹⁸ successfully identified seven taekwondo techniques. Hong et al.⁴⁸ utilized the joint detection of two RGB-D cameras to recognize sandbag kicking movements, which can effectively solve the blocking effect of sandbags on the human body. Its authors find that, in general, correctly detected joints have a longer accumulative geodesic distance (AGD) than that of incorrectly detected ones. The sum of AGDs for each camera was compared to select a reliable camera. Joint locations were determined by maximizing the sum of AGDs. Choi and Joo⁴⁹ also used two Kinect cameras and utilized a comparison of the accuracy of naive Bayes (NB), multilayer perceptron (MLP), and decision trees (DT) in identifying taekwondo poomsae. Bianco and Tisato⁵⁰ adopted a bottom-up approach: keyframes were first recognized by k-means and then aligned by DTW.

Dr. Hachaj et al. has devoted his research to pattern recognition and technical analysis in karate, and much of his related research collected data through the Kinect sensors.^{47,51–57} He proposed the gesture

description language (GDL). The idea of the GDL algorithm is to encode the gesture sequence as a series of static keyframes in a defined order. These sequences are encoded by a context-independent syntax called GDL scripts. The GDL script consists of a set of rules that together create a knowledge database, like an expert system. GDL does not require a large training dataset.^{54,55,58} Gestures are defined by the user in the GDL script. A user can use any number of physical features he or she needs in each rule definition. Gestures are split into keyframes and appear in a certain order within a given limit. He found that the accuracy of motion capture can limit the accuracy of GDL classifiers.^{54,55} To demonstrate that human motion can be modeled in low-dimensional feature space,⁵³ Dr. Hachaj attempted to reduce the dimensions of the skeletal features. K-means clustering was designed to perform clusterization of input data in feature space. The center of clusters together with the standard deviation of elements from a cluster and epsilon value creates the definition of GDL keyframes of actions. Subsequently, Idris et al.^{59,60} used GDL to recognize Silat actions. Hachaj et al.⁵² proposed a sampling strategy: 20 or 25 time-varying 3D human joint coordinates obtained by the Kinect V2 are preprocessed into a nine-dimensional angle-based time-varying feature set, a 15-dimensional or a 16-dimensional distance-based feature set. The data is resampled to a uniform length using cubic spline interpolation, after which each action is represented by 60 samples. Ultimately 540 (60×9), 900 (60×15), or 960 (60×16) dimensional variables are input to the classifier (SVM, ANN, Random Forest).

With the development of deep learning, the performance of pose estimation algorithms has greatly improved, and they have gradually been used for recognizing martial arts movements. In Wen^{61,62}, two-dimensional coordinates were obtained through OpenPose, and a video registration method based on DTW was proposed. Similarly, in Guo⁶³ and Zhang et al.,⁶⁴ HRNet is employed first, then DTW is used to register the video sequentially. Dong et al.⁶⁵ utilized AlphaPose to extract the two-dimensional coordinates of the human skeleton, and 15 tai chi actions were recognized by an LSTM with an attention module. The authors found that the recognition accuracy improved slightly with the introduction of speed and acceleration features. Li⁶⁶ argue that most current methods have not been able to adequately mine multiple spatiotemporal connections in skeletal data. Therefore, Li et al. proposed a spatiotemporal feature network to capture multiple spatiotemporal dependencies. The coordinate embedding structure mines the hidden spatial connections of the joints themselves by upscaling the skeletal data, the spatiotemporal feature encoder mines the spatiotemporal information between the joints in the same and different frames by encoding the skeletal data, and the network-based classifier learn the encoding map to finally output the classification results of Tai Chi gestures.^{66,67} Echeverria and Santos^{68–70} fused multimodal

data and designed a method to identify the action of a karate kumite: a heart rate sensor recording physiological data, two unmanned aerial vehicles taking videos, object detection and OpenPose extracting kinematic data.

Graph convolutional network (GCN) is very suitable for learning spatial topological structures. Spatiotemporal graph convolutional network (ST-GCN)⁷¹ and Actional-Structural Graph Convolutional Networks (ASGCN)⁷² are a kind of deep learning model combining GCN with temporal convolution, which can learn the spatiotemporal features of actions. Recently, Guo et al.^{73–75} used eight cameras to take karate kumite video synchronously, and mapped 2D human coordinates extracted by OpenPose back to 3D space. A new GCN algorithm improved by Delaunay triangulation increasing edges of the graph is proposed, which can recognize three actions: punch, kick and move. And the joint coordinates can be used to record the movement tracks of both players on the field. Using dual-stream GCN and ConceptNet, Guo also proposed a weakly supervised learning method based on domain knowledge maps to recognize these karate actions.⁷⁶ Wang⁷⁷ proposed two improved algorithms: one is ST-GCN enhanced by attention, the other is Bi-directional LSTM with two-stream input, and both of them can recognize eight Tai Chi actions. By considering the causal relationship of large joints driving small joints in human movement, Ye et al.^{78,79} proposed a Tai Chi action recognition method that integrates causality and a spatiotemporal graph convolutional network. They assigned edge weights to the skeletal graph based on causal relationships between joints and augmented the GCN with the weights as auxiliary information to increase the attention of the joints with higher drive in the neural network.

Wearable sensor-based methods

Table 2 records the research of martial arts action recognition based on wearable sensors. The common recording types of equipment are inertial sensors (accelerometer, gyroscope), earth magnetometers, infrared-reflective markers, and surface electromyography. The motion capture based on infrared-reflective markers has the highest accuracy, which is generally considered to be millimeter level. Tits et al.⁸⁰ proposed a feature extraction method that abandoned morphological features. While they paid attention to representing the spatial relationship between local and global motion: for the local joint trajectory, the coordinate system is established with the parent joint as the origin. For the global joint trajectory, the coordinate system was established with the center of the human body as the origin, and the feature's dimension was reduced by factor analysis. Cizmic et al.⁸¹ used data obtained from sensors embedded in smart boxing gloves to recognize punches and target objects. Motion features were further extracted from the angular velocity, speed,

Table 2. An overview of wearable-based martial arts action recognition.

Year	Data acquisition modality	Algorithms	Types of martial arts	Accuracy
2017 ⁸²	XSens MVN MoCap	Temporal feature pyramid, hidden conditional random field	Tai Chi	For 21 actions, the proposed method can reach 96.2%.
2018 ²⁰	Wearable Shadow2.0 (contains 17 inertial measurement units, including three-axis accelerometers, gyroscopes and magnetometers)	PCA, HMM	Karate	For eight types of kicks, the proposed method can reach 92.0%.
2020 ⁸³	Qualisys MoCap System	SVM	Karate	Not mentioned
2020 ⁸⁴	Arduino single-chip computer with three-axis accelerometer	ANN	Boxing	Not mentioned
2020 ⁸⁵	Inertial Measurement Unit (IMU)	One body joint: GMM, Fisher vector; SVM; Multiple body joints: DTW.	Wu style Tai Chi	Single-node method 98.33% (two-classification), 93.86% (ten-classification). Multi-nodes method 99.92% (two-classification), 90.45% (ten-classification). The proposed 1D CNN method reaches 99.74%.
2021 ⁸⁶	Inertial Measurement Unit (IMU)	PCA, KNN, SVM, NB, LR, DT, ANIN, Radial Basis Function Neural Network (RBFNN), 1D-CNN	Baduanjin	The proposed MLP method reaches 99.0%.
2021 ⁸⁷	Marker-based MoCap, electromyogram capture, pressure plate, RGB camera	PCA, MLP, KNN, NB, RF	Historical European martial arts (HEMA)	
2021 ^{88,89}	Electromyogram capture	MLP, LDA Classifier, DT	Yang style Tai Chi	99.51%.
2021 ⁹⁰	Three-axis accelerometer; three-axis gyro sensor	DTW, CNN	Taekwondo	CNNs reach 92.08% and 88.70% on boxing and Taekwondo.
2022 ⁹¹	IMU	FFT + SVM, FFT + KNN	Taekwondo	FFT + KNN reaches 96.8%.
2022 ⁹²	Electromyogram capture	XGBoost, SVM, NB, KNN, DT, RF	Tai Chi	The XGBoost reaches 88.17%.
2022 ⁹³	IMU	DTW + kNN	Baduanjin	83.03%.
2023 ⁹⁴	RGB camera + IMU	Two-stream CNN for fusing features called KarateNet	Karate	76.47%.
2023 ⁸¹	IMU sensor + force measurement unit	DT, RF, SVM, kNN, NB, MLP, LR	Boxing	The SVM reaches 93.03%.

acceleration, and force collected by the gloves to implement ML models for classifying various striking techniques, as well as distinguishing between hitting a valid target (e.g. punching bag, punching bag, glove) or an invalid target (e.g. concrete wall).

In Hachaj and Ogiela,²⁰ the joint coordinates were converted into eight angles, and the dimension was reduced by PCA to construct the HMM motion recognition model. Two different recognition methods are adopted by Xu et al.⁸⁵ In one approach, only one track of the node was used. The segmented track was clustered by using the GMM and Fisher vector, and then SVM is used as the classifier for recognition. Another approach utilized the trajectories of multiple nodes: using the DTW algorithm to calculate the minimum distance between two trajectories. Then the category of action was determined according to the similarity of trajectory. In Li et al.,⁸⁶ the time-domain features of the IMU signal are used: mean value, variance, standard deviation, skewness, kurtosis, and quartile deviation. Dharmmesta et al.⁹¹ classify kicks of taekwondo by SVM and KNN. The models that use kurtosis features achieve the best performance. Based on the data collected by IMUs, Li et al.⁹³ used a variety of sequence-based methods (DTW + SVM, DTW + KNN, HMM, and RNN) to recognize Baduanjin actions and estimate correctness. Yadav et al. considered that a single modality cannot contain all the information about the motion. They designed KarateNet,⁹⁴ a dual-stream convolutional neural network that could fuse RGB video and IMU data.

EMG can be used to assess muscle function and activity levels. By analyzing EMG signals, information about the strength, duration and coordination of muscle contractions can be obtained. Klempous et al.⁸⁷ collected data through motion capture based on reflective spheres, surface electromyography, pressure plates, and RGB cameras and classified the actions of armed martial arts. Ma et al.^{88,89} measured and characterized the surface electromyographic signals of the Yang style Tai Chi “brush knee and twist hip on both sides” action using the BP neural network, LDA classification, and decision tree, to reveal the basic patterns of Tai Chi movements and the differences between short-term and long-term practitioners. Faisal et al.⁹² employed classical machine learning techniques (SVM, Naive Bayes, KNN, DT, RF, XGBoost) to identify and evaluate the gait classification accuracy of Tai Chi masters and ordinary subjects in single-task and dual-task walking scenarios.

Optimization of pose estimation

Human pose estimation (HPE) is a computer vision task to predict the position of human joints. As a kind of contactless motion capture, it has high application value for activity understanding. But martial arts have many special postures, such as rolls, high kick (shown

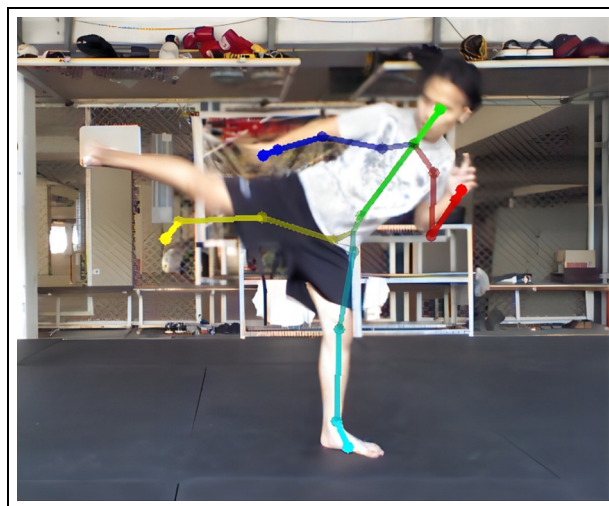


Figure 5. Pose estimation is difficult to apply well to special postures.⁹⁶ (Reproduced with permission, courtesy of IEEE.)

in Figure 5), handstands, tucks, and turns. These postures cause considerable challenges in estimating the human pose in martial arts. Moreover, occlusion in multiplayer scenes can interfere with the pose estimation model.^{33,95}

Using pose estimation for martial arts actions sometimes results in missing or incorrectly detected limb coordinates. Duvelek et al.⁹⁸ anomalously detected these cases by designing a set of rules. Yasin et al.⁹⁹ constructed a knowledge base containing a priori existing knowledge to infer missing or corrupted information in motion capture data. A kd-tree was then constructed in a parallel fashion on the GPU to efficiently search and retrieve this knowledge quickly from the knowledge base in the form of nearest neighbors, querying for motions with missing joints or markers. Finally, an objective function with multiple error terms recovers the 3D joint or marker trajectories in parallel on the GPU. In Alharbi et al.,¹⁰⁰ before the RGB-D information was fed into the Kinect's own pose estimation model, the noises of the depth map were reduced with an extended Kalman filter, which could reduce the misidentification of key points caused by cross-arm movements in 24-form Yang style Tai Chi. Baek and Kim¹⁰¹ used eight Kinect cameras to simultaneously record taekwondo players, retrieved joints in voxels synthesized from the depth map and used K-means to determine toe, ankle, and knee joint positions. Kamel et al.¹⁰² designed a Kinect depth-map-based pose estimation algorithm that did not use the 3D coordinates automatically generated by Kinect, but instead fed the depth map into a convolutional neural network to regress the 3D coordinates, and a weighted average of joint angle differences was used as a score to compare the user's movements to the 24 Tai Chi movements of six athletes. Wang⁷⁷ worked on solving the problem of key point misidentification and key point misconnection in OpenPose by sampling the human pose flow

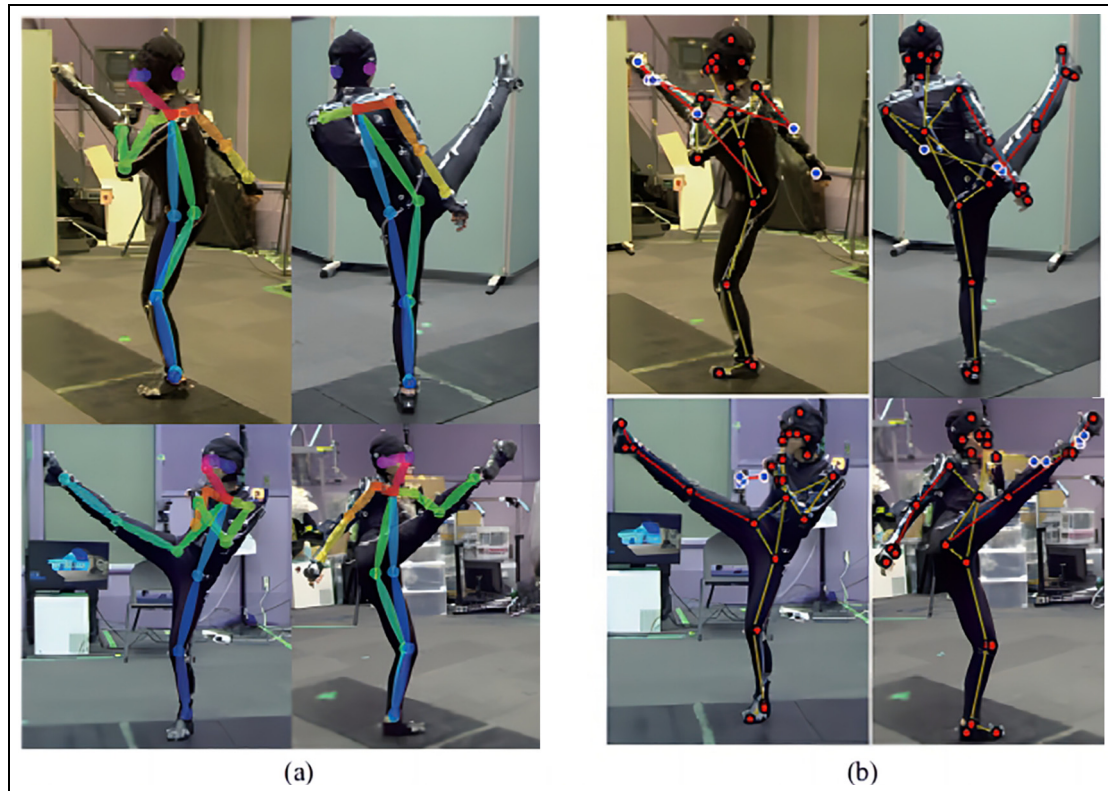


Figure 6. Transfer learning-based pose estimation results.⁹⁷ (Reproduced with permission, courtesy of IEEE.)

information in the image sequences to optimize the OpenPose. Guo⁶³ proposed a lightweight high-resolution pose estimation network HS-GatNet for capturing the motion information of Plum Blossom boxing, which improved the HRNet and took monocular camera video as input.

Frequent errors and physical inconsistencies in pose estimation can accumulate and lead to unnatural feature behavior. Peng et al.^{103,104} enhanced the reliability of pose estimation based on reinforcement learning. Motion imitation with reinforcement learning provides a powerful tool for learning processes to refine imperfect and noisy pose sequences, compensate for missing frames, and consider the physical constraints of the character and the environment by reproducing skills in physical simulations. Girón et al.¹⁰⁵ consider that the method of representing the human body representation as a graph (joints as nodes and limbs as edges) is more useful than the common methods which are estimated directly from the image. Because this method allows focus more on the relationship of body parts and thus on valuable information. Thereby, they utilize a semantic graph convolutional network to perform 3D pose estimation from a single image.

It is worth noting that, many state-of-the-art pose estimation models take COCO as their dataset, however, COCO lacks toe and heel annotations, which are essential in actual human motion analysis. This problem often causes inconvenience in practical applications (e.g. judging the quality of kicking, hook and trip movements). Training and opening some of these martial arts

pose estimation models to include toe and heel key points are critical to facilitate the wider application of non-contact martial arts motion capture. Ooke et al.⁹⁷ took Openpose and HRNet for transfer learning on a self-constructed taekwondo dataset (34 landmarks), and the motion capture's accuracy improved figure 6(b) and was significantly better than before figure 6(a). MADS¹⁷ is a dataset that can be used to research 3D human pose estimation. MADS includes many sports such as Tai Chi, Karate, Hip Hop, Jazz, Basketball, Volleyball, Soccer, Rugby, Tennis, and Badminton. The dataset also includes multi-view and stereo-view data, as well as synchronized and calibrated ground truth poses associated with each video. To evaluate the accuracy of the pose estimator on the MADS dataset, Le et al.¹⁰⁶ proposed a pose estimation measure that performs 2D pose estimation first and then compares it based on the length of joints, deviation angle of joints, and deviation distance of key points projected into 3D space. Le and Scherer tested the 3D pose estimation performance of RepNet, Mediapipe, Lifting from the Deep, and V2V PoseNet on the MADS dataset, and found that V2V PoseNet had the best estimation value for the deviation angle.¹⁰⁷ Le and Scherer further evaluated the performance of various action recognition and human segmentation algorithms on MADS.^{66,107} While performing the action recognition task, ST-GCN, DD-Net and PA-GesGCN achieve 100% accuracy in action recognition on MADS when the ratio of training set to test set is greater than 1.5. For human segmentation,

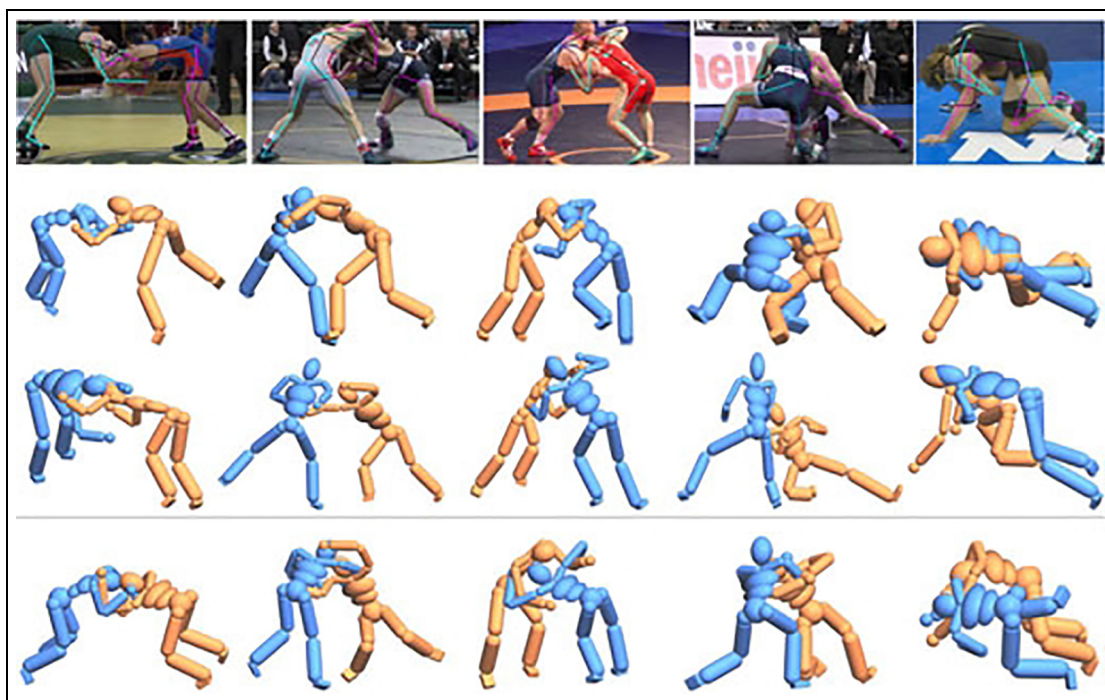


Figure 7. Generating closely interacting 3D pose-pairs from 2D annotations.¹⁰⁹ (Reproduced with permission, courtesy of IEEE.)

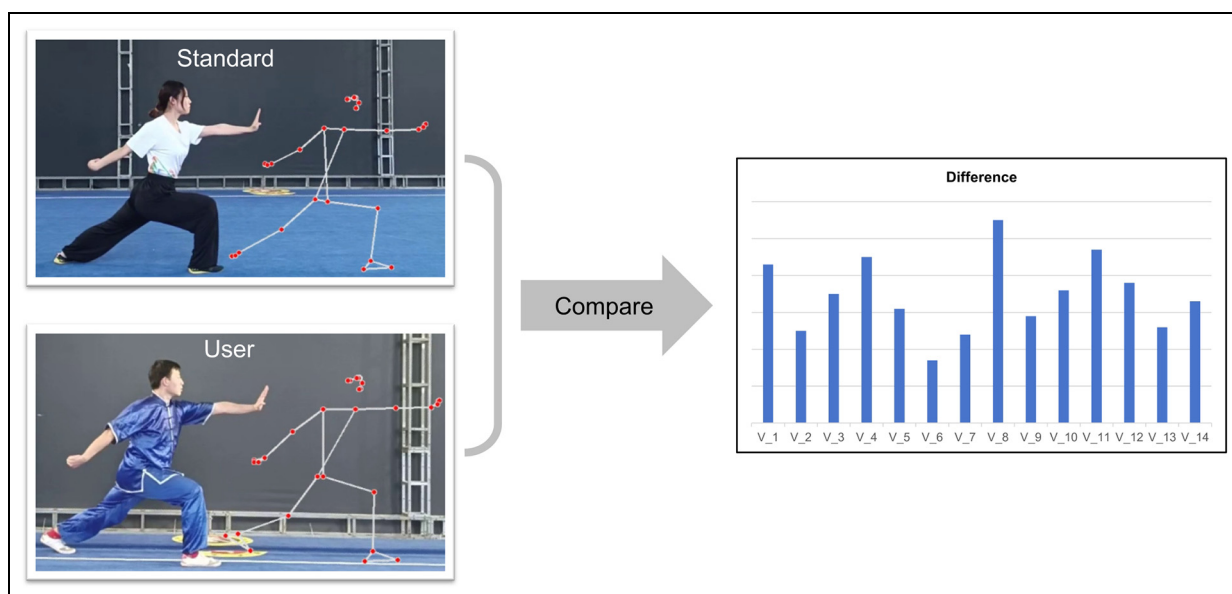


Figure 8. Evaluating actions by template method.

CenterMask achieved the best performance among Mask R-CNN, PointRend, TridentNet, TensorMask, and CenterMask.

As shown in Figure 7, if 3D coordinates can be obtained from 2D images, it will greatly reduce the cost of data collection. Ma et al. point out that 2D coordinates are not sufficient to provide enough information about the movement of martial arts.¹⁰⁸ Yin et al.¹⁰⁹ attempted to restore 3D pose pairs of two wrestlers from 2D poses.

Action evaluation

Template method

The template method involves using the standard action as a template and comparing the similarities and differences between the user's action and the template (as shown in Figure 8). The template method can be further categorized into direct comparison and distance metrics. Direct comparison: the joint angles of the participants at the same moment in time were made

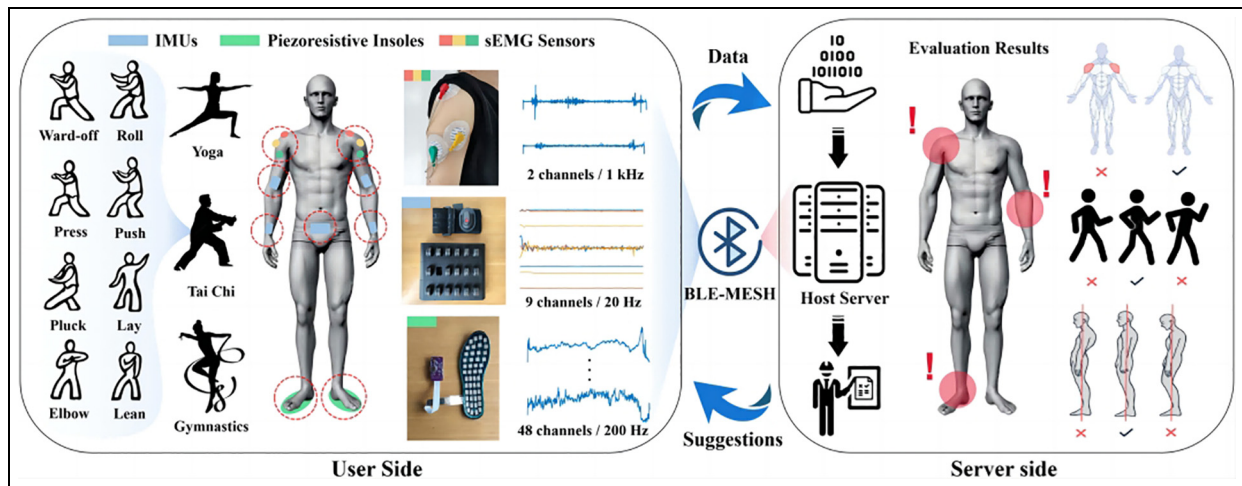


Figure 9. Recognize incorrect actions by AI.¹²¹ (Reproduced with permission, courtesy of IEEE.)

different from the standard joint angles one by one and these differences were used as guidance.^{63,110,111} This method, although it feels somewhat rigid, provides tangible guidance. Distance metric: comparing the similarity between the user's action and the standard action using different distance functions, for example, Euclidean distance^{112,113}, Cosine distance,^{114,115} Mahalanobis distance,¹¹⁶ canonical correlation analysis,¹¹⁷ and DTW.^{118–120}

Classification

As shown in Figure 14, the grading of action quality can be classified by machine learning. Pang et al.¹²² argue that the Yin and Yang of body movement in Tai Chi is a special movement pattern, and that head-shoulder and shoulder-hip should trend to move in opposite directions. So, the distance between head-shoulder and shoulder-hip is used as a feature to distinguish between correct or incorrect form. In this way, action evaluation becomes a signal classification. Li et al.⁸⁶ extracted the time-domain features of the signal: mean, variance, standard deviation, skewness, kurtosis, and interquartile deviation, for classifying the Baduanjin movements into three levels: good, pass, and fail. Zhong et al.¹²³ argue that valuable features are concentrated in a portion of the spectrum and useless features lead to the degradation of the SVM classifier's accuracy, speed, and efficiency. Therefore, a taekwondo action evaluation method that filters features based on the genetic algorithm was proposed. Wessa et al.¹²⁴ feed body coordinates into the MLP model to separate correct or incorrect kickboxing postures. Burdack et al.⁸³ trained SVM classifiers to identify unique movement patterns for individual athletes based on kinematic joint angle waveforms of karate front kicks at different levels of fatigue during training. Yamanaka et al. utilized DT¹²⁵ and CNN¹²⁶ to grade the "focus," "dynamic," "balance," and "total" of taekwondo actions from triaxial acceleration at the waist.

Quintero Ovalle et al.¹²⁷ argue that boxing with different parts of the hand could produce different sounds when hitting on the sandbag. They attempted to improve recognition by combining audio input with conventional inertial sensors: from each axis of the accelerometer and gyroscope, they obtained: mean, standard deviation, maximum, minimum, and zero crossings; from the audio signal, they extracted: mean, standard deviation, and the three largest Fourier transform peaks (frequency and phase) for input to the classifier (however the audio did not improve the accuracy due to the lack of expertise of the subjects).

Zhang and Zhou¹²⁸ classified the flying kicks of Nanquan into correct and incorrect, and incorrect movements identified by Bayesian algorithms. Emad et al.^{129,130} identified correct and incorrect karate moves by SVM, KNN, and DT. Hülsmann et al.¹³¹ classified incorrect movements in Tai Chi push hands, first using DTW alignment, then random forest (RF) to filter features, and finally an SVM classifier to find incorrect movements. Jang et al.¹³² used IMU to measure the joint motion data and impact force during kicks, and trained a CNN model to identify anomalous kicks that should not be scored in Taekwondo competitions. As shown in figure 9, Tang et al.¹²¹ proposed a Wearables-Based Multi-Column Neural Network to fusion physiological data recorded by multi-source sensors (EMG, piezoresistive insoles, IMU) to recognize 17 incorrect movements of Tai Chi practitioners.

Regression

This methodology entails constructing a regression model through the aggregation of diverse features linked to martial arts movements. By minimizing the prediction error, the model undergoes optimization, facilitating the precise estimation of the score for the martial arts movements. Subsequently, this refined model predicts the score by considering the amalgamated features, thus ensuring a meticulous assessment

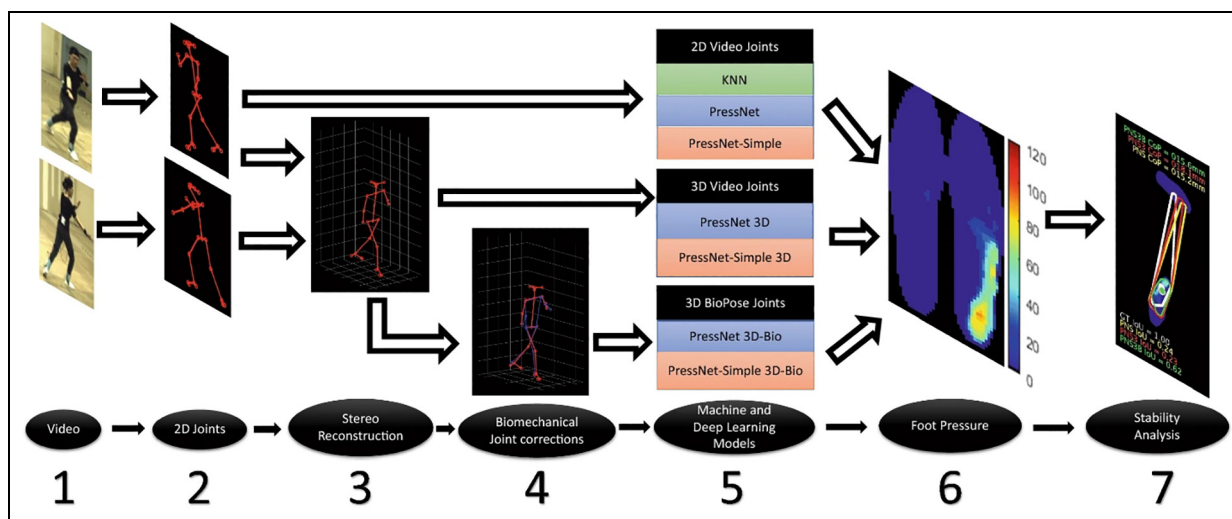


Figure 10. Learning dynamics from human pose.²² (Reproduced with permission, courtesy of Springer.)

of the martial arts move's performance. Tuong Thanh et al.¹³³ weighed the angular differences and summed them to obtain a score for the action. With the help of the Sigmoid function, the output of the SVM decision function is mapped to a posterior probability from 0 to 1. The magnitude of the probability is used to express the degree of similarity between the action to be evaluated and the standard action of a professional athlete.^{77,134}

In Bianco and Tisato,⁵⁰ a scoring method is proposed: a score representing the effectiveness and quality of the move is given as the output, which is independent of the length of the movement. The final score is obtained by regression between human judgment and the standardized DTW distance obtained by each technique. Dai¹³⁵ used a graph convolutional neural network for feature extraction of captured Tai Chi actions and a Siamese network for action evaluation. By comparing the Siamese network with the DTW-based method, Dai finds that the accuracy of the Siamese network is better than that of DTW, and the advantage of the Siamese network is more obvious when the user's movements are more deviated from the standard movements.

PressNet^{22,136} is the first algorithm based on capturing kinematic information regressing kinetics (center of pressure and support surface), with kinematic data recorded by marker-based MoCap and kinetic data recorded by plantar pressure insoles (shown in Figure 10).

Other applications

Serving elite athletes

AI has also been used to serve elite martial arts athletes. Athlete selection is the beginning of a competitive sports career, it is the foundational work of selecting children and adolescents with good athletic talent and competitive potential or reserves to participate in sports

training. Wiacek et al.¹³⁷ used a decision tree classifier to determine which of the following: wrestling, triple jump, badminton, and tennis the athletes are more suitable for based on human anthropometrics. Martínez et al.¹³⁸ employed ANN to determine the physical fitness level of taekwondo athletes according to body composition, balance control, anaerobic power, and isokinetic strength, which can provide a basis for the evaluation of training effectiveness and athlete selection. AI can provide technical and tactical data mining and knowledge discovery for adversarial matches. Soo et al.¹³⁹ utilized logistic regression and conditional interference classification tree (shown in Figure 11) to explore the key fighting characteristics and performance metrics associated with match success. The analysis shows that "total takedown success," "defending takedown success" and "catch success" had the greatest impact on explaining the results of the Elite Pencak Silat Tournament. An outcome depends not only on the performance of athletes and opponents in the match but also on the act of judgment. Holmes et al.¹⁴⁰ used a Bayesian hierarchical model to determine whether MMA matches were controversial or not, to reduce the potential bias of referees, and to enhance the fairness of the matches.

Sports betting

Predicting the winner of a match is a topic of interest for sports betting. Hitkul et al.²³ trained six classifiers (MLP, RF, DT, stochastic gradient descent classifier, SVM and KNN) based on data such as past performance of fighters, body shape and other data publicly available in UFC stats,¹⁴¹ and correlation analysis was performed on the metrics. McQuaide²⁴ analyzed the metrics according to the importance of each variable in the gradient boosting decision tree (GBDT) UFC winner prediction model. Hachaj et al.¹⁴² proposed a video annotation method for karate group hands (sparring)

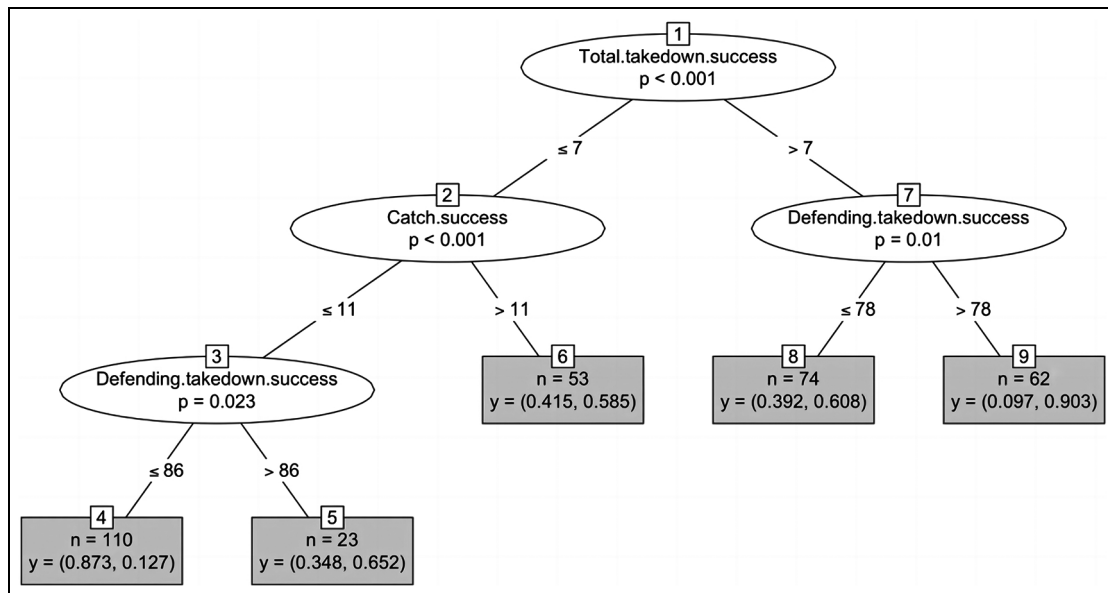


Figure 11. Conditional interference classification tree highlighting the probability of wins and losses.¹³⁹ (Reproduced with permission, courtesy of Taylor & Francis.)

that allows both quantitative (numerical) and qualitative (categorical) features to be computed. PCA and KNN were used to construct classifiers to predict wins and losses. Holmes et al.¹⁴³ used Markov chains to model a match as an actual battle. The proposed Markov model achieves an accuracy of 61.77%, not only surpassing the Bradley-Terry model (54.13%) and logistic regression (47.71%), but even achieving comparable results to the bookmakers only with smaller data.

Health care

Martial arts can be employed as a means of exercise intervention. Grzenda et al.¹⁴⁴ take SVM, RF, and logistic regression to predict the therapeutic effect of Tai Chi on depression, respectively. Meng et al.¹⁴⁵ predict the health status of frail older adults after an intervention via multiple machine learning models. Wu et al.¹⁴⁶ analyzed and interpreted the data using random forests and demonstrated that Tai Chi can alleviate symptoms of chronic fatigue syndrome by strengthening brain network connections and that long-term aerobic exercise improves sleep quality in patients with chronic fatigue syndrome. In Guo et al.,¹⁴⁷ kinematic data was obtained by accelerometers and energy expenditure of the Baduanjin exercise with a portable metabolic analyzer, and an ANN model was trained to predict the energy expenditure of the Baduanjin exercise.

Yet the head injuries caused by combat sports should not be ignored. Zhan et al.¹⁴⁸ used MLP to predict Entire Brain Strain in MMA fighters. Falcone et al.¹⁴⁹ extracted acoustic features from fighters' speech and identified mild traumatic brain injuries with SVM. Brain deformation is capable of detecting traumatic brain injury in clinical applications.¹⁵⁰ A random forest classifier with the spectral density of linear acceleration

and angular velocity was developed to classify head impact types (e.g. soccer, car accidents, mixed martial arts) by Zhan et al.¹⁵¹ Zhan et al.¹⁵² further proposed to combine unsupervised domain adaptation and deep neural networks to predict whole-brain strains and strain rates. The proposed method significantly outperforms not only the baseline model without domain adaptation but also other domain adaptive methods in terms of prediction accuracy.

Animation generation

Interactively synthesizing novel combinations and variations of character movements from different movement skills is a key issue in computer animation. Starke et al.¹⁵³ proposed a deep learning framework that generates new martial arts movements from raw motion capture data in a controllable way (as shown in Figure 12), and the system can be used for both offline and online movement generation, providing an intuitive workflow integration with the animator workflow interface. Won et al.¹⁵⁴ proposed a deep reinforcement learning-based framework for the generation of two-player interactions, where an intelligent body can autonomously learn boxing and fencing match actions in a physics simulator. Zhou et al.¹⁵⁵ argue that errors accumulate and feed back into the network over time when using autoregressive techniques. When these accumulated errors reach a certain level, it will lead to unnatural or abnormal phenomena in the generated animation. To address this problem, Zhou et al. proposed a method called auto-conditioned Recurrent Neural Network for synthesizing highly complex human movements while maintaining fluidity and naturalness. Li et al.¹⁵⁶ proposed a neural network model

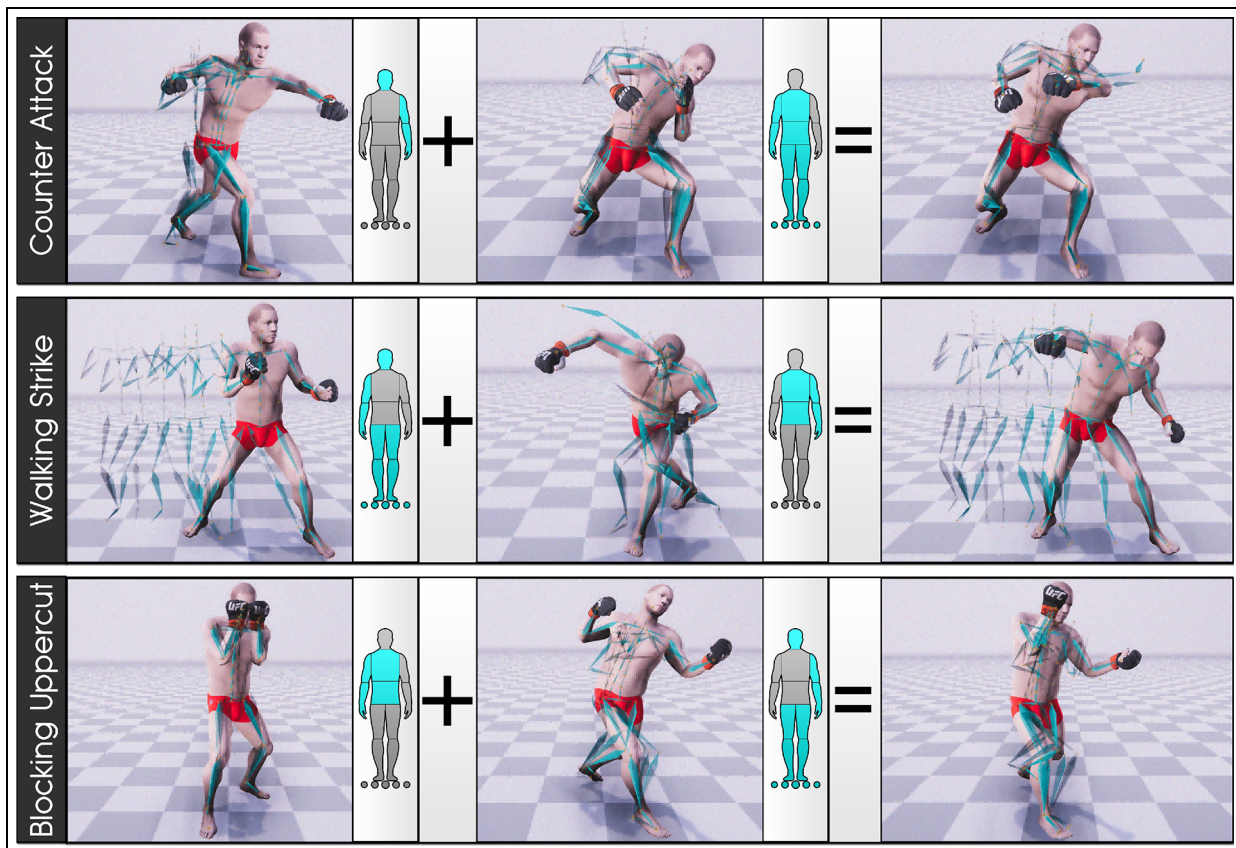


Figure 12. Generate new martial arts movements from motion capture data.¹⁵³ (Reproduced with permission, courtesy of the author.)

based on a spatial-temporal transformer for solving the glide problem in complex animations. The model utilizes a deep learning technique that combines keyframes and foot contact constraint information to synthesize high-quality animations. The spatial-temporal transformer captures and models complex dynamic patterns, including joint motion, gait periodicity, and more. This way allows the model to better understand the patterns of motion in the animation, leading to more accurate animation prediction and generation.

Competition field segmentation

Sun et al.¹⁵⁷ were devoted to segmenting the competition field of the Wushu routine, and the field image is modeled by the Gaussian mixture model. The expectation maximization (EM) algorithm is used to estimate the unknown parameters of the GMM model.

Discussion

Based on the investigation of the literature on action recognition, pose estimation and action evaluation, we summarize a paradigm that is expected to provide inspiration for potential researchers and promote their work. The paradigm reflects the technical process that

can be used in the intelligent martial arts training system (shown in Figure 13). The data collected are usually kinematic data (acceleration, motion trajectory), video frame sequences (RGB flow, depth map flow), and physiological data (EMG). For video frame sequences, image features can be extracted not only by manually designed methods (optical flow, Fischer vectors, dense trajectories, gradient histograms) or by automated methods (CNN), but also by human pose estimation to transform video frame sequences into skeleton information. Kinematic data can be transformed into kinematic features (angle, distance), time domain features (mean, variance, standard deviation), and frequency domain features (highest magnitude, total energy, highest magnitude located, entropy, kurtosis, skewness).^{93,158} When dealing with sequences of different lengths both zero-padding¹⁵⁹ and alignment¹⁶⁰ can be utilized, noise or jitter can be reduced by smoothing algorithms,^{161–163} and PCA can be used to reduce dimensionality for high-dimensional data.^{88,164} Image features and kinematic features can be used individually¹⁶⁵ or jointly¹²¹ as inputs for action recognition and action evaluation. Figure 13 stimulates thinking about the most appropriate scenarios for individual AI algorithms in applications. Many studies have dealt with action recognition, so the status of action recognition in AI + martial arts

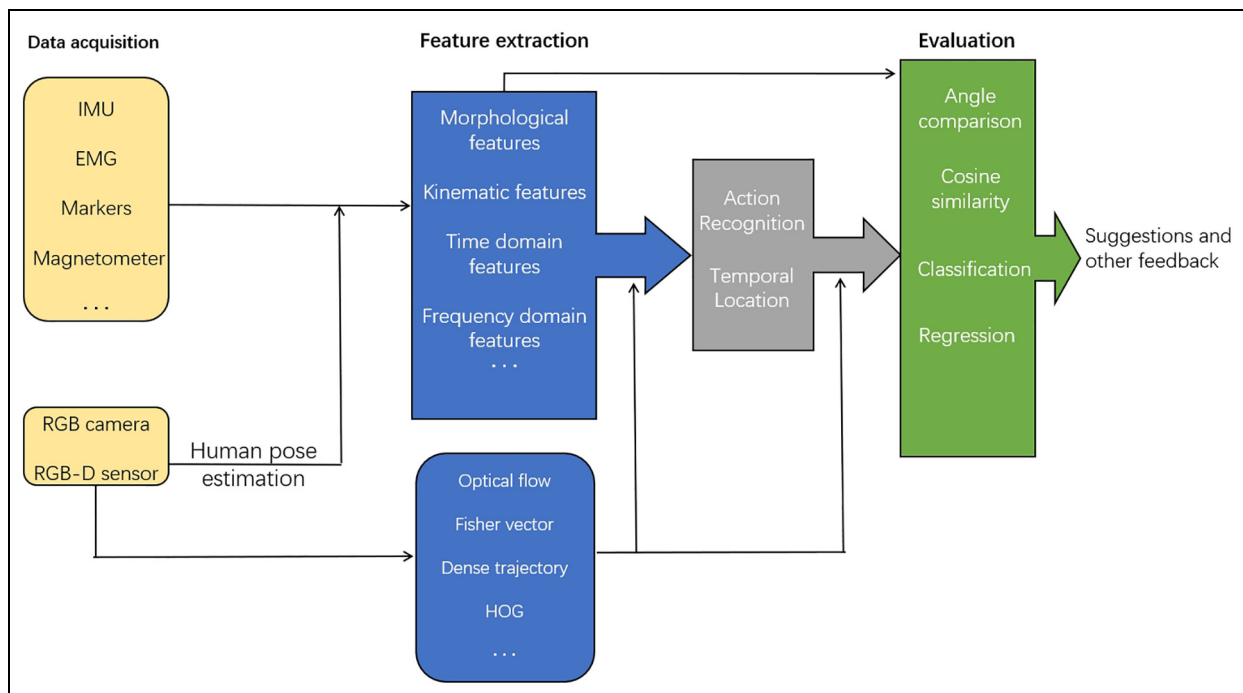


Figure 13. Potential technical process for an intelligent martial arts training system.

is discussed. In addition, potential approaches for developing intelligent martial arts systems based on these techniques are explored. Finally, the prospect of future research is discussed.

Suitable scenarios for AI algorithms

In articles concerning the application of AI techniques for martial arts, an in-depth discussion of the basic features, pros, and cons of these techniques, as well as their optimal applicability in different scenarios, can help researchers to better select appropriate algorithms and to explore other possible alternatives.

Machine learning.

- (1) K-Nearest Neighbors (KNN) is a basic supervised algorithm that makes classification or regression predictions by finding the closest K training samples in the feature space. KNN's advantage is that it does not require a training phase and is suitable for small-scale data and nonlinear problems. However, for large-scale data, KNN is less efficient and susceptible to noisy data. It can be employed for small datasets and problems that do not require a lot of preprocessing, especially for classification problems, for example, classifying Taekwondo⁹¹ and freestyle wrestling.¹⁶⁶
- (2) K-Means is an unsupervised clustering algorithm that uses clustering to divide the data into K clusters and optimize the objective function by minimizing the distance within each cluster. The advantage of K-means is that it can classify

martial arts moves into different clusters by measuring the distance between samples.⁵³ However, the K-means algorithm requires a predetermined value of the number of clusters K, which is sometimes unknown or difficult to determine. The K-means algorithm is sensitive to noise and outliers, and may assign them to different clusters by treating them as outliers.

- (3) Support Vector Machines (SVM) is a supervised algorithm for classification and regression, and is used to find an optimal hyperplane to maximize the marginal distance between categories. The SVM's advantage is that it can effectively handle high-dimensional data, does not require much feature engineering, and is still effective when the data dimension is high. SVM is popularly used for classifying martial arts data.^{23,83,128,167} However, SVM is sensitive to noises and must choose the appropriate kernel function. SVM not only performs well when dealing with low-dimensional data¹⁶⁸ but can also model high-dimensional data.¹²⁹
- (4) Tree-Based Algorithms/ decision tree (DT) which is a basic ML algorithm that makes decisions based on tree structure. A decision tree learns based on instances, dividing them by testing their attributes and constructing new nodes. Each node represents a judgment condition on an attribute, each branch represents a possible attribute value, and each leaf node represents a classification result. Decision trees have the advantages of being intuitive and easy to understand, and insensitive to some missing data and outliers. On the basis of DT, models such as Random Forest (RF),^{158,169}

Gradient Boosting Decision Tree (GBDT),²⁴ and eXtreme Gradient Boosting (XGBoost)⁹² have been developed. Random forest is an integrated learning model consisting of multiple decision trees with higher prediction accuracy and better noise immunity. GBDT is an integrated learning model that optimizes the loss function by iteratively adding weak learners. XGBoost is an efficient GBDT implementation that uses a gradient-boosting algorithm with better generalization and higher computational efficiency. In addition, tree-based algorithms have shown good performance in data mining for martial arts.^{23,145}

- (5) Hidden Markov Models (HMM) is a probabilistic graphical algorithm for sequence modeling, often applied to time series data analysis. Because of its ability to capture hidden states and state transitions in sequences, HMM is well-suited for modeling sequence data, for example, using HMM to classify the movements of the Baduanjin⁹³ and taekwondo.¹⁷⁰ However, domain knowledge is typically required to define the states and state transitions of the HMM, and, when dealing with long sequence data, the modeling efficiency of the HMM is weak.
- (6) Dynamic Time Warping (DTW) is a method for measuring the similarity between two sequences, which is usually used for time series data alignment and matching. DTW can handle sequences of different lengths, and it is suitable for time series alignment,^{93,171} action recognition,^{64,172,173} and action similarity comparison.¹¹⁸ The drawback is that DTW has high computational complexity. Even when the improved FastDTW is used to process a large amount of data and high-dimensional sequences, its time delay is still relatively long.

Deep learning.

- (1) Multilayer Perceptron (MLP) is a type of artificial neural network with a very simple structure. The multilayer structure of MLP allows it to have a strong expressive ability to learn complex features from the original data. MLP is suitable for dealing with nonlinear problems and highly complex tasks. MLP can implement classification^{174,175} and regression¹⁴⁷ problems for martial arts data, and it has a better generalization ability. MLP has been used for action recognition^{176,177} and winner prediction.¹⁷⁸ However, it has been found that for data in tabular form, tree-based models usually achieve better results than MLP.¹⁷⁹
- (2) Convolutional Neural Networks (CNNs) can process data in a variety of modalities, such as images,¹⁸⁰ skeletons,¹⁸¹ and data collected by IMUs.⁹⁰ The convolutional layer of CNN effectively captures the local features of an image and can be used to extract both spatial^{182,183} and temporal^{184,185} features. The CNNs allow the

researchers to flexibly modify the structure to fulfill their purpose.¹⁸⁶ The inference process in CNNs tends to be more complex and difficult to interpret compared to traditional machine learning methods, which makes people have less trust in the models.

- (3) Long Short-Term Memory (LSTM) Network is a variant of Recurrent Neural Network (RNN) used to process sequential data with long-term memory and the ability to capture long-distance dependencies, for example, using LSTM to learn temporal relationships from CNN-extracted feature maps of martial arts movements¹¹⁵ or skeleton sequence.^{187,188} Yet LSTM models are relatively complex and often require larger datasets to feed the model.
- (4) Graph Neural Network (GNN) is a kind of neural network specialized in modeling data with graph structure. GNN can make full use of the structural information of the graph, to extract the features of the nodes better and improve the accuracy of prediction. A graph convolutional neural network (GCN) combining GNN and CNN can be used to model the skeleton sequences of martial arts players.^{189,190} GNNs tend to have a larger number of parameters than other neural network models, and therefore require more computational resources and time for training and inference.

The status of action recognition in AI + martial arts

Action recognition is a crucial task in enabling computers to comprehend human actions and appropriately assign labels to each action. As previously expounded, a substantial amount of research has been devoted to the development of action recognition algorithms. In the context of AI and martial arts, action recognition holds immense significance. The deployment of action recognition algorithms in martial arts instruction, training, and competition scoring can be highly effective. Specifically, the categorization of diverse movements in martial arts through action recognition can facilitate training monitoring of the athletes, simulation of tournaments, and provision of teaching assistance to coaches.

The accurate recognition of user movements is a prerequisite for realizing movement quality assessment in intelligent martial arts systems. Moreover, the techniques employed for pre-processing, feature extraction, and modeling of human motion data for action recognition can be adapted for action quality evaluation with appropriate modifications. As such, action recognition is a fundamental component for achieving intelligent martial arts systems and serves as a crucial pre-requisite for action quality assessment.

Online short video platforms like “TikTok” have emerged as a popular medium for the dissemination of martial arts due to their extensive user base and swift dissemination channels. The complex nature of online

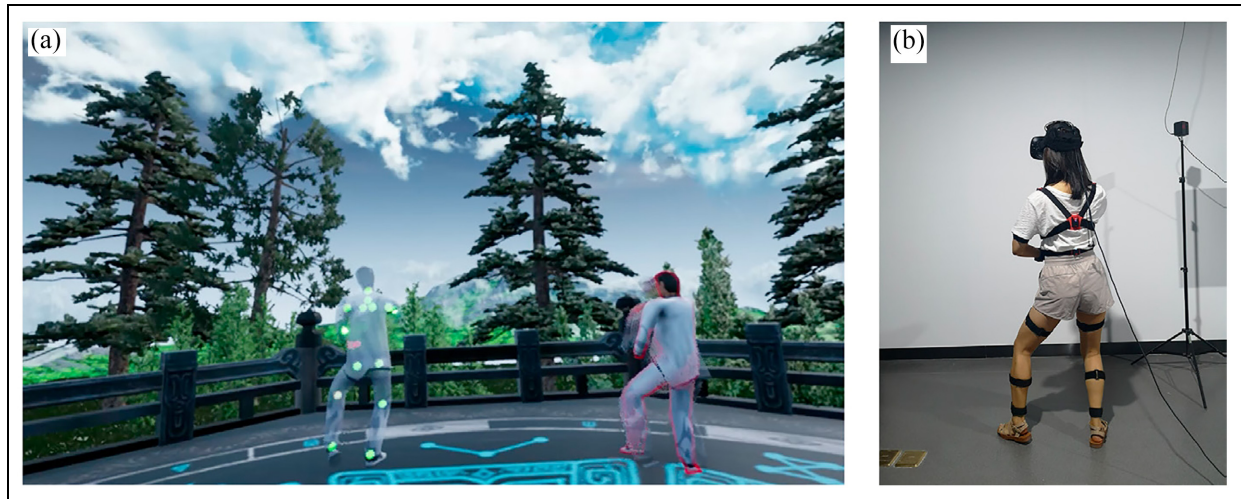


Figure 14. Tai Chi teaching system: (a) The coach and avatar in virtual scenes and (b) the learner in real world.¹¹³ (Reproduced with permission, courtesy of Elsevier.)

video content necessitates the use of recommendation algorithms to recommend videos tailored to user preferences. However, the process often requires extensive manual annotation for video categorization and tagging, with the latter providing a more detailed description of the video content.¹⁵⁹ Employing action recognition algorithms can help automate the process of classifying and labeling videos, detecting highlight clips,³⁴ and even generating video summaries,^{191,192} which facilitates recommendation and retrieval¹⁹³ by the platform, thus significantly reducing labor costs.

In summary, action recognition plays a vital role in the intersection of AI and martial arts, facilitating improvements in the technical proficiency and training outcomes of martial arts. The integration of intelligent technology with martial arts is a significant step toward promoting the digital and intelligent transformation of martial arts, with far-reaching positive implications.

Intelligent martial arts system

Intelligent martial arts teaching/training system. An intelligent martial arts teaching system has the potential to mitigate the shortage of teachers, augment the enjoyment of teaching, and facilitate the widespread dissemination of martial arts. The implementation of an intelligent martial arts teaching system requires careful consideration of the following aspects:

- (1) **Development of a comprehensive data collection and analysis system:** This system should be designed to record and analyze students' movement techniques (as Figure 14(b)), such as data on movement trajectories, speed, and strength. The resulting data can then be effectively processed to provide valuable feedback and suggestions to both teachers and students.

- (2) **Design of an intelligent teaching system:** Utilizing advanced technologies such as AI, computer vision, and machine learning, this system should evaluate and differentiate students' movement techniques, providing targeted learning guidance¹¹³ and even suggestions in alignment with specific teaching objectives.
- (3) **Creating Interactive Interfaces:** The system should create interactive interfaces for the instructor and the students,^{194,195} and in addition to the student interfaces used for movement training (as shown in Figure 14(a)), it can also include the instructor interfaces for monitoring and adjusting student training.
- (4) **Development of a multimedia learning resource library:** To enrich students' martial arts knowledge and cultural literacy, a diverse range of learning resources should be provided, including movement videos, martial arts knowledge materials, and introductions to martial arts culture.¹⁶¹

Realizing an intelligent martial arts teaching system requires a combination of technological tools and resource libraries to provide quality teaching services. At the same time, the role of the teacher remains crucial to improve students' learning effectiveness and actual combat skill level by assessing and guiding their movement techniques. As for the intelligent martial arts training system, it needs to identify and evaluate martial arts training in a targeted manner and provide personalized training suggestions and feedback to martial arts learners, such as incorporating domain knowledge to statistics on a user's skills (as shown in Figure 15). The system should also provide coaches with scientific teaching assistance through intelligent monitoring and recording to improve the teaching effect and quality, thereby promoting the digitization, intelligence, and modernization of martial

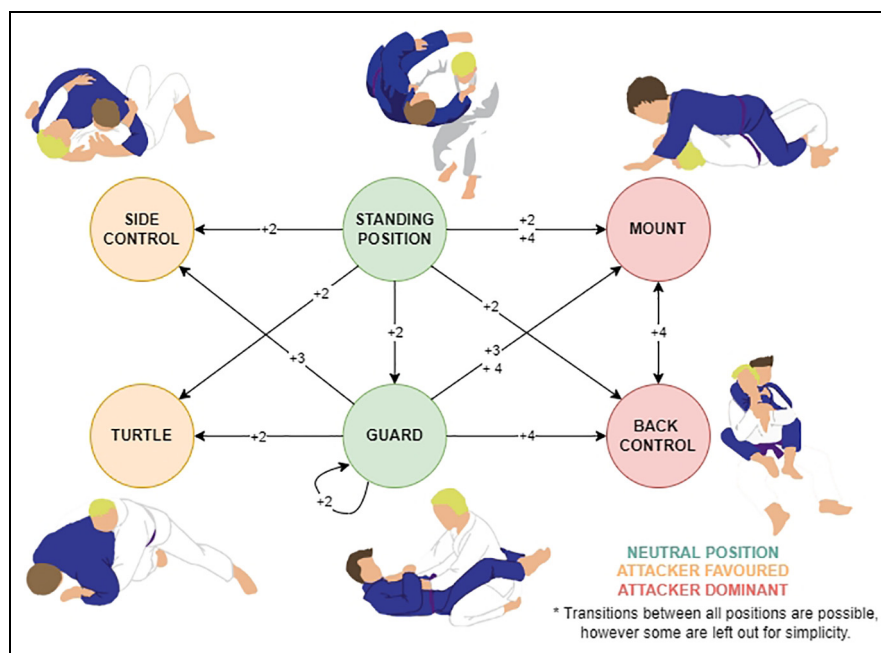


Figure 15. The scoring rules of jiu-jitsu combat positions training system.³³ (Reproduced with permission, courtesy of the author.)

arts sports training. Compared with the intelligent teaching system, the intelligent training system must also include monitoring of both internal and external load to effectively gauge athletes' physical exertion and optimize their training regimen accordingly.

Intelligent analysis system for combat competition techniques and tactics. The construction of an intelligent analysis system for fighting competition techniques and tactics holds great potential for improving the competitive level of martial arts. Through technical means, this system can quickly and effectively identify and analyze athletes and events, providing precise technical and tactical guidance to coaches and athletes. Furthermore, the intelligent analysis system can enhance match visualization, improve judging accuracy, and reduce the influence of human factors on match results, ensuring the fairness, openness, and transparency of matches. Thus, the development of an intelligent technical and tactical fighting competition analysis system is of critical importance to the growth and improvement of martial arts competitions and sports. The realization of an intelligent technical and tactical fighting competition analysis system entails the integration of diverse technical tools, establishment of an efficient data collection, analysis, and processing mechanism, and utilization of advanced methods such as deep learning based on big data to enhance data analysis capabilities and provide coaches and athletes with precise analysis and guidance support. The system can be achieved by leveraging various technical tools, such as AI, computer vision, and big data technology. Figure 16 illustrates a technical and tactical intelligence analysis system. Specifically:

An intelligent technical and tactical fighting competition analysis system requires the integration of various technical means, the establishment of an efficient data collection, analysis, and processing mechanism, and methods such as deep learning based on big data to improve data analysis capabilities and provide coaches and athletes with accurate analysis and guidance support.

Effective data collection and processing of the competition site, such as athletes' action data and competition videos, are crucial for the basis of the intelligent analysis system. However, occlusion between fighters in a fighting competition poses a challenge for intelligent technology. Transient occlusion scenes, such as athletes' heads, chests, or waists being less exposed, can affect the accuracy and objectivity of the analysis system, resulting in scoring bias and inaccurate scoring results. Therefore, an overhead shot can be considered to identify and effectively solve the occlusion problem accurately.¹⁹⁷

In addition, action recognition and analysis using computer vision technology are necessary to identify competition athletes' movements,¹⁶⁹ displacements, and postures,^{162,198} and mining and analysis of competition data in combination with algorithmic models. For example, the inter-boxer distance can be calculated for understanding the fluctuations in proximity between boxers as a function of various actions and behaviors.¹⁹⁶ Based on the results of data analysis, analysis models applicable to different game types of techniques and tactics should be developed, and statistical analysis of different athletes, opponents, and game sessions, should be conducted. The combination of machine learning and analysis models can provide coaches and athletes with accurate suggestions and guidance.

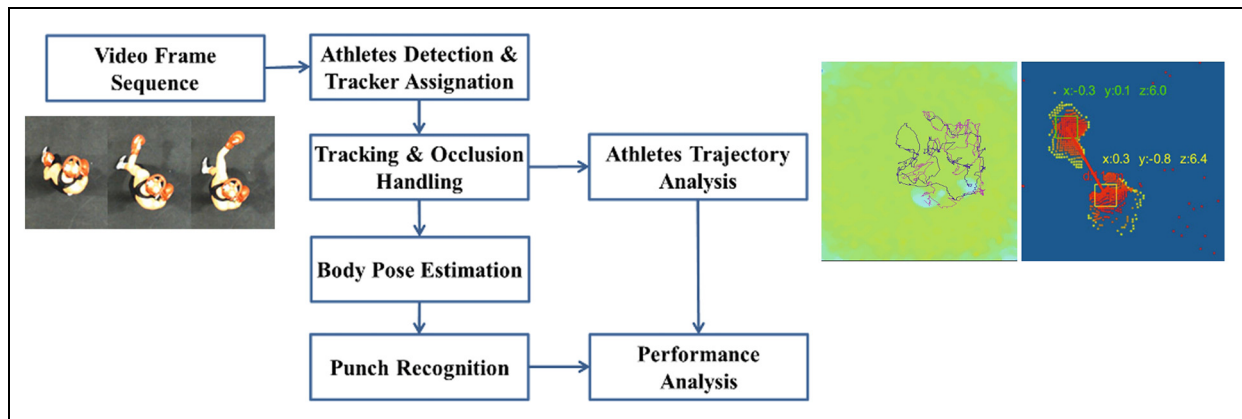


Figure 16. Non-invasive performance measurement in combat sports.¹⁹⁶ (Reproduced with permission, courtesy of Springer.)

Finally, a high-quality interactive interface and professional database are required to improve the response speed and analysis quality of the analysis system,¹⁹⁶ while further improving the efficiency and precision of analysis technology and data processing.^{199–201}

Martial arts routine intelligent judging system. The implementation of an intelligent scoring system for martial arts routines has significant implications for improving the efficiency and accuracy of scoring in martial arts competitions.⁶⁴ This system can provide fast and precise scoring results, thereby enhancing the fairness and transparency of the competitions. Furthermore, it can establish more scientific and equitable scoring standards, reducing disputes among competitors and coaches, and ensuring impartiality in the competition. Moreover, constructing an intelligent scoring system can promote the digitization and intelligence of traditional martial arts, boost the development and popularization of martial arts, and enhance the inheritance and development of martial arts as a cultural heritage.¹⁵⁹ The adoption of an intelligent scoring system for martial arts routines can elevate the technical level and overall quality of martial arts competitions, increase the status and influence of martial arts in the global competitive system, and advance the modernization and internationalization of martial arts.⁶⁴

Martial arts routines consist of a series of choreographed movements performed in a specific order. The key frames of each action can be determined using temporal action positioning, which requires accurately identifying the start and end times of each action.^{160,202} To evaluate an athlete's score, a combination of movement normality¹¹¹ and rehearsal level²⁰³ should be considered. Kinematic parameters such as joint angle, distance, and trajectory can be used to assess movement normality, while the similarity metric based on machine learning and deep learning can be used to evaluate rehearsal levels (shown in Figure 17).

Future perspective for AI + martial arts

Innovation of data acquisition methods. Current contactless motion information acquisition is restricted to a single RGB camera or an RGB-D sensor, which is limited in accuracy and can hardly be used for biomechanical analysis like wearable motion capture. Multi-view 3D pose estimation²⁰⁴ (Figure 18) and 3D human reconstruction algorithm²⁰⁵ are expected to capture more accurate motion information and richer details which has stronger robustness for adversarial sports such as Sanda, Kumite, and wrestling. Single-view 3D pose estimation and single-view 3D human reconstruction (Figure 19) also have good prospects to be used under equipment limitations, at lower cost and are more conducive to replication.²⁰⁶

Few-shot learning for individualized modeling. Although few-shot learning is currently being used primarily for sports video recognition,²⁰⁷ techniques like data augmentation¹⁹³ and leave-one-out validation,²⁰⁸ which can reduce the dependence on sample size, have been successfully applied to machine learning for sports data. The range of dissemination and the number of practitioners vary greatly among martial arts styles. For some types of martial arts with few practitioners, it is difficult to collect sufficient high-quality data. Few-shot learning is indispensable, and the value of this method for the conservation of endangered martial arts should be emphasized. In addition, the concept of few-shot learning can be used to analyze the similarities and differences among high-level athletes.

Multimodality and cross-modality help to discover new tasks. Modeling between motion information, speech, mechanics, physiology, and neurological data is expected to facilitate smarter analysis of martial arts movements and to promote the health functions of martial arts. In other sports, this idea has been successfully used in areas such as fatigue monitoring,²⁰⁹ biomechanical analysis,²¹⁰ injury prevention and treatment.²¹¹

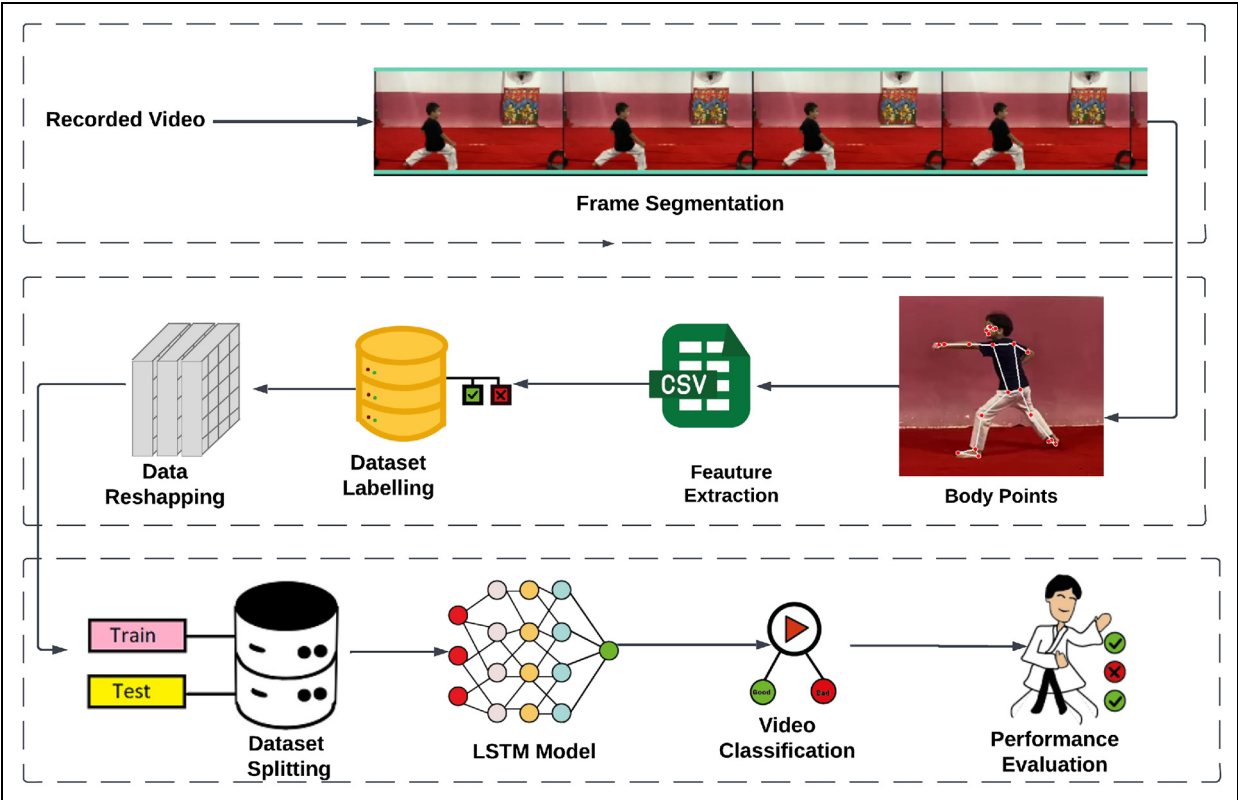


Figure 17. Karate kata scoring system.²⁰³ (Reproduced with permission, courtesy of IEEE.)

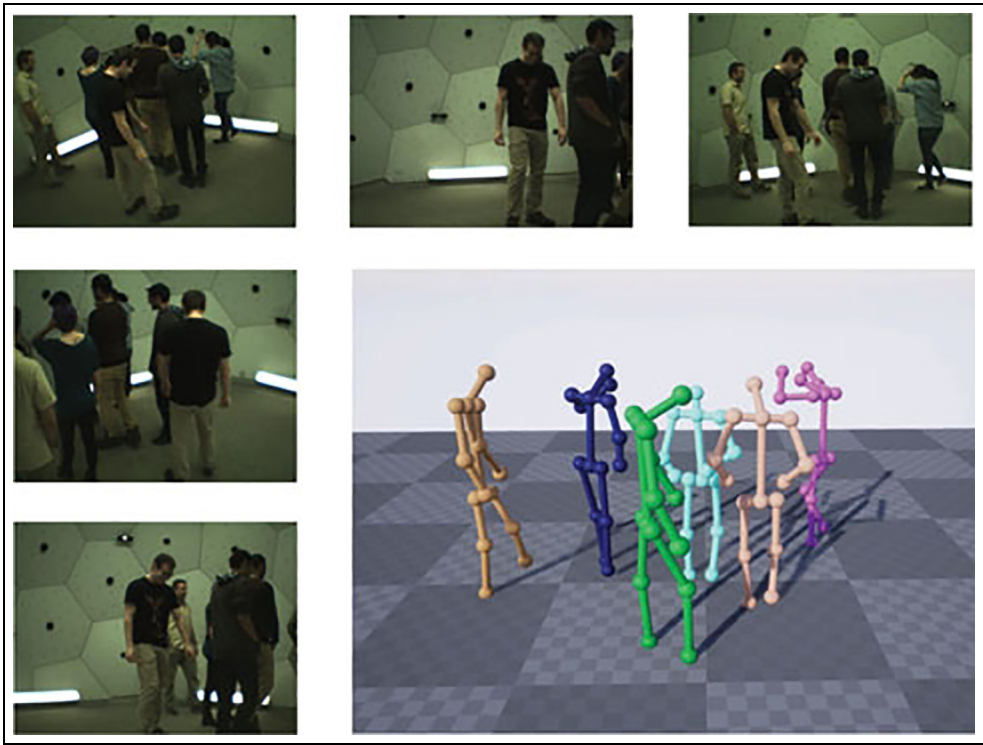


Figure 18. Multi-person 3D pose estimation from multiple views.²⁰⁴ (Reproduced with permission, courtesy of IEEE.)

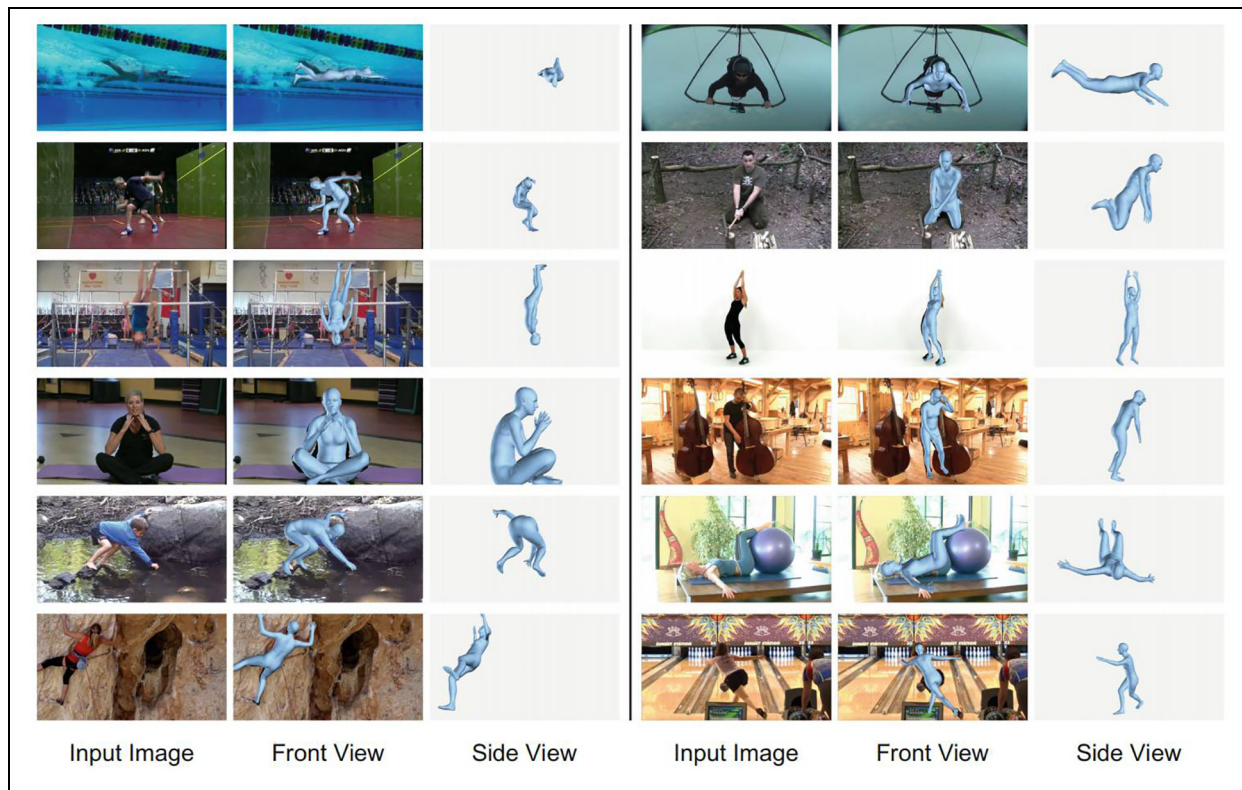


Figure 19. Single-view 3D pose estimation and human reconstruction.²⁰⁶ (Reproduced with permission, courtesy of Springer.)

Explainable AI-enabled martial arts. The widespread use of AI also brings with it the problem of the “black box.” As many complex AI systems use highly optimized algorithms, their decision-making processes are often opaque and difficult to understand, making it difficult for people to trust and accept these systems. Therefore, to solve this problem, there is a need to develop explainable AI to improve the transparency and understandability of AI decisions. For example, to assess the interpretability of established AI models for basketball prediction models, Wang et al.²¹² utilize a model agnostic interpretation method based on local interpretability to derive and assess the exact reasoning behind the model predictions. Martial arts are about internal and external integration. For martial arts routines, their scoring can be categorized into the normality of the moves (external) and the essence (internal). AI can not only score external movements but also can evaluate the essence, vital energy, spirit and style of routines based on explainable models, which may be the next issue worth exploring. The development of explainable techniques for intelligent analysis of techniques and tactics is also promising for adversarial combat sports.

Conclusions

The applications of AI in martial arts are surveyed in this paper. Most researchers have taken widely spread martial arts such as Tai Chi, Karate, and Taekwondo as the subject of their research.

These applications cover a wide range of specific disciplines, including not only action recognition, pose estimation, and action evaluation, but also serving elite athletes, sports betting, health care, animation generation, and competition field segmentation.

Methods for action evaluation can be grouped into three categories: template, classification, and regression. In general, the template method does not require a large training sample and has good interpretability, the classification method can give a rating of the movement and the regression method can give a specific score.

Earlier research relied more on the manually designed features. Feature engineering such as corner detection, optical flow, time domain features, frequency domain features, and dimensionality reduction can be a good way to improve the performance of recognition algorithms. With the development of deep learning, since the emergence of OpenPose, the performance of pose estimation models has been greatly improved, and the human skeleton is increasingly used for the recognition and evaluation of martial arts actions. In recent years, deep learning models such as LSTM and GCN have been used more frequently for temporal or spatial modeling of martial arts actions.

AI can assist in martial arts teaching, training and competition. Most research is dedicated to improving the accuracy of the algorithms, and there is a lack of research in the evaluation of movements. It should be

clear to researchers that good work should reflect the properties of the martial arts themselves, rather than adapting algorithms and models to them.

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Supplemental material

Supplemental material for this article is available online.

References

1. Martial art — definition, history, types, facts, <https://www.britannica.com/sports/martial-art> (accessed 10 November 2022).
2. Kung fu — definition, history, types, facts, <https://www.britannica.com/sports/kung-fu> (accessed 10 November 2022).
3. Theeboom M and Knop PD. Asian martial arts and approaches of instruction in physical education. *Eur J Phys Educ* 1999; 4(2): 146–161.
4. Martial arts, https://en.wikipedia.org/wiki/Martial_arts (accessed 10 November 2022).
5. Martinková I and Parry J. Martial categories: clarification and classification. *J Philos Sport* 2016; 43(1): 143–162.
6. Charles M and Terry F MD. The martial arts. *J Philos Sport* 2006; 17(3): 645–676.
7. Angleman AJ, Shinzato Y, Van Hasselt VB, et al. Traditional martial arts versus modern self-defense training for women: some comments. *Aggress Violent Behav* 2009; 14(2): 89–93.
8. Douris P, Chinan A, Gomez M, et al. Fitness levels of middle aged martial art practitioners. *Br J Sports Med* 2004; 38(2): 143–147. discussion 147.
9. Tsang TW, Kohn M, Chow CM, et al. Health benefits of kung fu: a systematic review. *J Sports Sci* 2008; 26(12): 1249–1267.
10. Bu B, Haijun H, Yong L, et al. Effects of martial arts on health status: a systematic review. *J Evid Based Med* 2010; 3(4): 205–219.
11. Origua Rios S, Marks J, Estevan I, et al. Health benefits of hard martial arts in adults: a systematic review. *J Sports Sci* 2018; 36(14): 1614–1622.
12. Yuqing Z, Mingliang C, Haoyang Z, et al. VR technology and application in martial arts. In: *2021 IEEE 7th international conference on virtual reality (ICVR)*, pp.240–245. DOI: 10.1109/ICVR51878.2021.9483700.
13. Pang Y, Wang Q, Zhang C, et al. Analysis of computer vision applied in martial arts. In: *2022 2nd international conference on consumer electronics and computer engineering (ICCECE)*, pp.191–196. DOI: 10.1109/ICCECE54139.2022.9712803.
14. Ke Y. Application of artificial intelligence in large-scale international wushu (taolu) events. In: *2021 2nd international conference on big data & artificial intelligence & software engineering (ICBASE)*, pp.575–578. DOI: 10.1109/ICBASE53849.2021.00113.
15. Zhang D. Research on the innovative development of Chinese mixed martial arts (MMA) events based on artificial intelligence. In: *2021 3rd international conference on machine learning, big data and business intelligence (MLBDBI)*, pp.785–788. DOI: 10.1109/MLBDBI54094.2021.00153.
16. Tits M, Laraba S, Caulier E, et al. Umons-taichi: a multi-modal motion capture dataset of expertise in Taijiquan gestures. *Data Brief* 2018; 19: 1214–1221.
17. Zhang W, Liu Z, Zhou L, et al. Martial arts, dancing and sports dataset: a challenging stereo and multi-view dataset for 3d human pose estimation. *Image Vis Comput* 2017; 61: 22–39.
18. de Souza Vicente C, Nascimento E, Emery L, et al. High performance moves recognition and sequence segmentation based on key poses filtering. In: *2016 IEEE winter conference on applications of computer vision (WACV)*, pp.1–8. IEEE.
19. Lee J and Jung H. Tuhad: taekwondo unit technique human action dataset with key frame-based cnn action recognition. *Sensors* 2020; 20(17): 4871.
20. Hachaj T and Ogiela M. Classification of karate kicks with hidden Markov models classifier and angle-based features. In: *2018 11th international congress on image and signal processing, biomedical engineering and informatics (CISP-BMEI)*, pp.1–5. IEEE.
21. Hachaj T, Ogiela M and Piekarczyk M. The open online repository of karate motion capture data: a tool for scientists and sport educators. In: *2017 IEEE symposium series on computational intelligence (SSCI)*, pp.1–5. IEEE.
22. Scott J, Ravichandran B, Funk C, et al. From image to stability: learning dynamics from human pose. In: *European conference on computer vision*, pp.536–554. Springer.
23. Hitkul KA, Yadav N and Dwivedy M. A comparative study of machine learning algorithms for prior prediction of ufc fights. In: Yadav N, Yadav A, Bansal J, et al. (eds) *Harmony search and nature inspired optimization algorithms advances in intelligent systems and computing*. Gurgaon: Springer Singapore, 2018, pp. 67–76.

24. McQuaide M. Applying machine learning algorithms to predict ufc fight outcomes, <https://cs229.stanford.edu/proj2019aut/> (accessed 30 March 2022).
25. Niewiadomski R, Kolykhalova K, Piana S, et al. Analysis of movement quality in full-body physical activities. *ACM Trans Interact Intell Syst* 2019; 9(1): 1–20.
26. Wagner T, Jäger J, Wolff V, et al. A machine learning driven approach for multivariate timeseries classification of box punches using smartwatch accelerometer sensor-data. In: *2019 Innovations in Intelligent Systems and Applications Conference (ASYU)*. pp. 1–6. DOI:10.1109/ASYU48272.2019.8946422.
27. Chen X, Chen Z, Li Y, et al. Immertai: immersive motion learning in VR environments. *J Vis Commun Image Represent* 2019; 58: 416–427.
28. Cho D, Lee H, Lee W, et al. Detecting anomalous kicks in taekwondo with spatial and temporal features. *IEEE Access* 2021; 9: 164928–164934.
29. Siarohin A, Lathuilière S, Tulyakov S, et al. First order motion model for image animation. *Advances in Neural Information Processing Systems* 2019; 32.
30. Szczesna A, Błaszczyshyn M and Pawlyta M. Optical motion capture dataset of selected techniques in beginner and advanced kyokushin karate athletes. *Sci Data* 2021; 8(1): 13.
31. Khasanshin I and Osipov A. Using an artificial neural network to develop an optimal model of straight punch in boxing and training in punch techniques based on this model and real-time feedback. *PLoS One* 2021; 16(11): 1–17.
32. Xu L, Wang Q, Lin X, et al. Skeleton-based tai chi action segmentation using trajectory primitives and content. *Neural Comput Appl* 2023; 35(13): 9549–9566.
33. Hudovernik V and Skocaj D. Video-based detection of combat positions and automatic scoring in jiu-jitsu. In: *Proceedings of the 5th international ACM workshop on multimedia content analysis in sports. MMSports '22*, pp.55–63. New York, NY: Association for Computing Machinery. ISBN 9781450394888. DOI: 10.1145/3552437.3555707.
34. Rodrigo M, Cuevas C, Berjón D, et al. Automatic highlight detection in videos of martial arts tricking. *Multimed Tools Appl* 2023; 83(6): 17109–17133.
35. Sun S, Wang F, Liang Q, et al. Taichi: a fine-grained action recognition dataset. In: *Proceedings of the 2017 ACM on international conference on multimedia retrieval. ICMR '17*, pp.429–433. New York, NY: Association for Computing Machinery. DOI: 10.1145/3078971.3079039.
36. Wang H, Kläser A, Schmid C, et al. Dense trajectories and motion boundary descriptors for action recognition. *Int J Comput Vis* 2013; 103(1): 60–79.
37. Hoshino K, Noguchi Y, Nakai Y, et al. Predicting the types of striking and thrusting motions by using deep learning. In: *Proceedings of the 2019 6th international conference on biomedical and bioinformatics engineering, ICBBE '19*, pp.199–202. New York, NY: Association for Computing Machinery. ISBN 9781450372992. DOI: 10.1145/3375923.3375954.
38. Wu E and Koike H. Futurepose -mixed reality martial arts training using real-time 3D human pose forecasting with a rgb camera. In: *2019 IEEE winter conference on applications of computer vision (WACV)*, pp.1384–1392. DOI: 10.1109/WACV.2019.00152.
39. Hörhan M and Eidenberger H. New content-based features for the distinction of violent videos and martial arts. In: *2013 IEEE international conference on acoustics, speech and signal processing*, pp.1836–1839. DOI: 10.1109/ICASSP.2013.6637970.
40. Mishra A and Srinivasa G. Automated detection of fighting styles using localized action features. In: *2018 2nd international conference on inventive systems and control (ICISC)*, pp.1385–1389. DOI: 10.1109/ICISC.2018.8399036.
41. Wang J. Research on wushu movement decomposition method based on image recognition. *Mod Electron Tech* 2017; 40(15): 33–36.
42. Sun R, Sun Y, Duan L, et al. Taijiquan key action recognition algorithm based on perceptron network. *China New Telecommunications* 2021, in Chinese.
43. Poulose A, Kim J and Han D. Motion classifier using deep learning architecture. In: *The 2nd Korea artificial intelligence conference*, pp.16–18.
44. Kong Y, Wei Z and Huang S. Automatic analysis of complex athlete techniques in broadcast taekwondo video. *Multimed Tools Appl* 2018; 77(11): 13643–13660.
45. Zhang H, Guo H, Liang C, et al. Grading tai chi performance in competition with rgbd sensors. In: Agapito L, Bronstein MM and Rother C (eds) *Computer vision - ECCV 2014 workshops*, pp.3–13. Cham: Springer International Publishing, 2014.
46. Salazar K, Garcia J, Mateus A, et al. Autonomous recognition of martial arts forms using rgb-d cameras. In: *2017 Congreso Internacional de Innovación y Tendencias en Ingeniería (CONITI)*, pp.1–5. IEEE.
47. Hachaj T, Ogiela M and Koptyra K. Application of assistive computer vision methods to oyama karate techniques recognition. *Symmetry* 2015; 7(4): 1670–1698.
48. Hong S, Kim M and Kim Y. Kick recognition system using dual rgb-d cameras. In: *Proceedings of the 2018 the 2nd international conference on video and image processing*, pp.12–16.
49. Choi CH and Joo HJ. Motion recognition technology based remote taekwondo poomsae evaluation system. *Multimed Tools Appl* 2016; 75(21): 13135–13148.
50. Bianco S and Tisato F. Karate moves recognition from skeletal motion. In: Baskurt AM and Sitnik R (eds) *Three-dimensional image processing (3DIP) and applications 2013*, vol. 8650, pp.86500K. Burlingame, California: International Society for Optics and Photonics, 2013, pp. 86500K1–86500K10.
51. Hachaj T. Key frames detection in motion capture recordings using machine learning approaches. In: *International conference on image processing and communications*, pp.79–86. Springer.
52. Hachaj T, Ogiela MR and Koptyra K. Human actions recognition from motion capture recordings using signal resampling and pattern recognition methods. *Ann Oper Res* 2018; 265(2): 223–239.
53. Hachaj T, Ogiela M and Koptyra K. Human actions modelling and recognition in low-dimensional feature space. In: *2015 10th international conference on broadband and wireless computing, communication and applications (BWCCA)*, pp.247–254. IEEE.
54. Hachaj T, Ogiela M and Piekarczyk M. Dependence of kinect sensors number and position on gestures recognition with gesture description language semantic classifier. In: *2013 federated conference on computer science and information systems*, pp. 571–575. IEEE.

55. Hachaj T and Ogiela M. Computer karate trainer in tasks of personal and homeland security defense. In: *International conference on availability, reliability, and security*, pp.430–441. Springer.
56. Hachaj T, Ogiela M and Koptyra K. Application of hidden Markov models and gesture description language classifiers to oyama karate techniques recognition. In: *2015 9th international conference on innovative mobile and internet services in ubiquitous computing*, pp.160–165. IEEE.
57. Hachaj T, Ogiela M and Koptyra K. Effectiveness comparison of kinect and kinect 2 for recognition of oyama karate techniques. In: *2015 18th international conference on network-based information systems*, pp.332–337. IEEE.
58. Hachaj T and Ogiela M. Computer system prototype for qualitative and quantitative evaluation of selected movement activities, pp. 73–76. DOI: 10.1145/3332305.3332309.
59. Idris W, Rafi A and Bidin A. Evaluating extrinsic feedback of martial art techniques using minimal values of rules in frame-based rules ranges. In: *2018 IEEE conference on systems, process and control (ICSPC)*, pp.112–117. DOI: 10.1109/SPC.2018.8703983.
60. Idris WMRW, Rafi A, Bidin A, et al. Developing new robust motion templates of martial art techniques using r-gdl approach: a case study of sscm. *Int J Arts Technol* 2019; 11(1): 36.
61. Wen X. *Research on the registration of Tai Chi video clips*. Master's Thesis, 2019, in Chinese. DOI: 10.27426/d.cnki.gxtd.2019.000570.
62. Ye S and Wen X. Taichi video registration method based on joint angle and dtw. *Comput Technol Autom* 2020; 39(01): 117–122.
63. Guo H. *Research and implementation of action training system on key point detection*. Master's Thesis, 2021, in Chinese. DOI: 10.27398/d.cnki.gxalu.2021.000248.
64. Zhang K, Pang Y, Wang Q, et al. Action recognition and temporal localization for wushu routine competition: taking the international long fist routine v3 as an example. In: *Proceedings of the 14th international conference on educational technology and computers*. DOI: 10.1145/3572549.3572643.
65. Dong L, Li D, Li S, et al. Tai chi action recognition based on structural lstm with attention module. In: *2019 international conference on image and video processing, and artificial intelligence*, vol. 11321, p. 1132110. International Society for Optics and Photonics.
66. Li X. *Research on skeletal gesture recognition based on a spatio-temporal feature network*. Master's Thesis, 2022, in Chinese. DOI: 10.27670/d.cnki.gcqdu.2022.002973.
67. Li X, Lia J, Liu L, et al. Recognizing skeleton-based hand gestures by a spatio-temporal network. In: Dong Y, Kourtellis N and Hammer B (eds) *Machine learning and knowledge discovery in databases. Applied data science track*, pp.151–167. Cham: Springer International Publishing, 2021.
68. Echeverria J and Santos O. Punch anticipation in a karate combat with computer vision. In: *Adjunct proceedings of the 29th ACM conference on user modeling, adaptation and personalization*, pp. 61–67.
69. Echeverria J and Kumitron CSO. Artificial intelligence system to monitor karate fights that synchronize aerial images with physiological and inertial signals. In: *26th international conference on intelligent user interfaces-companion*, pp.37–39.
70. Echeverria J and Santos O. Kumitron: A multimodal psychomotor intelligent learning system to provide personalized support when training karate combats. maied'21 workshop. In: *The first international workshop on multi-modal artificial intelligence in education*.
71. Yan S, Xiong Y and Lin D. Spatial temporal graph convolutional networks for skeleton-based action recognition. In: *Proceedings of the AAAI conference on artificial intelligence*, vol. 32. DOI: 10.1609/aaai.v32i1.12328.
72. Li M, Chen S, Chen X, et al. Actional-structural graph convolutional networks for skeleton-based action recognition. In: *2019 IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pp.3590–3598. DOI: 10.1109/CVPR.2019.00371.
73. Guo J, Li X, Liu J, et al. The application of behavior recognition technology based on graph convolutional network in karate tactics analysis. *J Nat Sci Human Norm Univ* 2021; 44(18): 81–87.
74. Liu J. *Research on key algorithms in tactical intelligence analysis system of karate competition video*. Master's Thesis, 2020, in Chinese. DOI: 10.27137/d.cnki.ghusu.2020.000419.
75. Guo J, Liu H, Li X, et al. An attention enhanced spatial-temporal graph convolutional LSTM network for action recognition in karate. *Appl Sci* 2021; 11(18): 8641.
76. Xie J, Li X, Zhou H, et al. Weak supervised behavior recognition algorithm based on domain knowledge graph. In: *2021 4th international conference on advanced electronic materials, computers and software engineering (AEMCSE)*, pp.899–904. IEEE.
77. Wang Z. *Human motion evaluation method based on multimodal information*. Master's Thesis, 2020, in Chinese. DOI: 10.27075/d.cnki.ghzdc.2020.000125.
78. Zhou Y. *Research on long sequence action recognition and prediction*. Master's Thesis, 2021, in Chinese. DOI: 10.27426/d.cnki.gxtd.2021.001534.
79. Ye S, Zhou Y, Fan H, et al. Joint learning of causality and spatio-temporal graph convolutional network for skeleton-based action recognition. *Comput Sci* 2021; 48: 130–135. DOI: 10.11896/jsjx.201200205
80. Tits M, Tilmanne J and Dutoit T. Morphology independent feature engineering in motion capture database for gesture evaluation. In: *Proceedings of the 4th international conference on movement computing*. MOCO '17. New York, NY: Association for Computing Machinery. ISBN 9781450352093. DOI: 10.1145/3077981.3078037.
81. Cizmici D, Hoelbling D, Baranyi R, et al. Smart Boxing Glove “RD α ”: imu combined with force sensor for highly accurate technique and target recognition using machine learning. *Appl Sci* 2023; 13(16): 9073.
82. Lin S, Lin Y, Chen C, et al. Learning and inferring human actions with temporal pyramid features based on conditional random fields. In: *2017 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, pp. 2617–2621. IEEE.
83. Burdack J, Horst F, Aragonés D, et al. Fatigue-related and timescale-dependent changes in individual movement patterns identified using support vector machine. *Front Psychol* 2020; 11: 2273.
84. Shan W, Fang Z, Wang J, et al. Design of intelligent boxing gloves based on arduino and neural network. *Journal of Fuyang Normal University (Natural Science)* 2020; 37(03): 54–58.

85. Xu L, Wang Q, Yuan L, et al. Using trajectory features for tai chi action recognition. In: *2020 IEEE international instrumentation and measurement technology conference (I2MTC)*.
86. Li H, Yap HJ and Khoo S. Motion classification and features recognition of a traditional Chinese sport (Baduan-jin) using sampled-based methods. *Appl Sci* 2021; 11(16): 7630.
87. Klempous R, Kluwak K, Atsushi I, et al. Neural networks classification for training of five German long-sword mastercuts-a novel application of motion capture: Analysis of performance of sword fencing in the historical European martial arts (hema) domain. In: *2021 IEEE 21st international symposium on computational intelligence and informatics (CINTI)*, pp.000137–000142. IEEE.
88. Guo W. *Research on surface EMG measurement and characteristics of typical movements of traditional Taijiquan*. Master's Thesis, 2021, in Chinese. DOI: 10.27312/d.cnki.gshsu.2022.000531.
89. Ma L, Guo W, Sun H, et al. Study on surface electromyography characteristics of traditional taijiquan typical movement. *J Optoelectron* 2021; 32: 257–265.
90. Amerineni R, Gupta L, Steadman N, et al. Fusion models for generalized classification of multi-axial human movement: validation in sport performance. *Sensors* 2021; 21(24): 8409.
91. Dharmmesta R, Jaya I, Rizal A, et al. Classification of foot kicks in taekwondo using svm (support vector machine) and knn (k-nearest neighbors) algorithms. In: *2022 IEEE international conference on industry 4.0, artificial intelligence, and communications technology (IAICT)*, pp.36–41. DOI: 10.1109/IAICT55358.2022.9887475.
92. Faisal MAA, Chowdhury MEH, Khandakar A, et al. An investigation to study the effects of tai chi on human gait dynamics using classical machine learning. *Comput Biol Med* 2022; 142: 1–14.
93. Li H, Khoo S and Yap HJ. Implementation of sequence-based classification methods for motion assessment and recognition in a traditional Chinese sport (baduanjin). *Int J Environ Res Public Health* 2022; 19(3): 1744.
94. Yadav S, Deshmukh A, Gonela R, et al. Ms-kard: a benchmark for multimodal karate action recognition. In: *2022 international joint conference on neural networks (IJCNN)*, pp.1–8. DOI: 10.1109/IJCNN55064.2022.9892646.
95. Rhodin H, Constantin V, Katircioglu I, et al. Neural scene decomposition for multi-person motion capture. In: *2019 IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pp.7695–7705. DOI: 10.1109/CVPR.2019.00789.
96. Kaewplee K, Khamsemanan N and Nattee C. A rule-based approach for improving kinect skeletal tracking system with an application on standard muay thai maneuvers. In: *2014 Joint 7th international conference on soft computing and intelligent systems (SCIS) and 15th international symposium on advanced intelligent systems (ISIS)*, pp.281–285. DOI: 10.1109/SCIS-ISIS.2014.7044763.
97. Ooke N, Ikegami Y, Yamamoto K, et al. Transfer learning of deep neural network human pose estimator by domain-specific data for video motion capturing. In: *2022 IEEE international conference on advanced robotics and its social impacts (ARSO)*, pp.1–6. DOI: 10.1109/ARSO54254.2022.9802965.
98. Duvelek S, Hölbling D, Baranyi R, et al. 2trax3: raising accessibility and everyday use of automatic motion analysis in (combat) sports via ml enhanced 2d to 3d estimation algorithms. In: *11th international conference on sport sciences research and technology support*, pp.128–135. DOI: 10.5220/0012165200003587.
99. Yasin H, Ghani S and Krüger B. An effective and efficient approach for 3d recovery of human motion capture data. *Sensors* 2023; 23(7): 3664.
100. Alharbi N, Liang Y and Wu D. A data preprocessing technique for gesture recognition based on extended-kalman-filter. In: *2017 IEEE/ACM international conference on connected health: applications, systems and engineering technologies (CHASE)*, pp.77–83. IEEE.
101. Baek S and Kim M. User pose estimation based on multiple depth sensors. In: *SIGGRAPH Asia 2017 posters*. 2017, pp.1–2.
102. Kamel A, Liu B, Li P, et al. An investigation of 3d human pose estimation for learning tai chi: a human factor perspective. *Int J Hum Comput Interact* 2019; 35(4-5): 427–439.
103. Peng XB, Kanazawa A, Malik J, et al. Sfv: reinforcement learning of physical skills from videos. *ACM Trans Graph* 2018; 37(6): 1–14.
104. Peng XB, Ma Z, Abbeel P, et al. Amp: adversarial motion priors for stylized physics-based character control. *ACM Trans Graph* 2021; 40(4): 1–20.
105. Girón V, Chau J, alfaro alfaró A, et al. 3D human pose estimation for martial arts analysis through graph convolutional networks, p. 43. DOI: 10.1117/12.2623512.
106. Le V, Nguyen T, Tran N, et al. Openpose's evaluation in the video traditional martial arts presentation. In: *2019 19th international symposium on communications and information technologies (ISCIT)*, pp.76–81. DOI: 10.1109/ISCIT.2019.8905243.
107. Le VH and Scherer R. Human segmentation and tracking survey on masks for mads dataset. *Sensors* 2021; 21(24): 8397.
108. Ma J, Ma L, Ruan W, et al. A wushu posture recognition system based on mediapipe. In: *2022 2nd international conference on information technology and contemporary sports (TCS)*, pp.10–13. DOI: 10.1109/TCS56119.2022.9918744.
109. Yin K, Huang H, Ho ESL, et al. A sampling approach to generating closely interacting 3d pose-pairs from 2d annotations. *IEEE Trans Vis Comput Graph* 2019; 25(6): 2217–2227.
110. Liu Y, Xu S, Song S, et al. Recognition and comparison of traditional wushu movements based on openpose. *Electron Compon Inf Technol* 2021; 5(3): 126–128.
111. Jayasekara S, Weerasinghe S, Abayawardana D, et al. Kaizen: computer vision based interactive karate training platform. In: *TENCON 2022 -2022 IEEE region 10 conference (TENCON)*, pp.1–6. DOI: 10.1109/TENCON55691.2022.9977691.
112. Chen H, Ma L, Lin Z, et al. Application of artificial intelligence vision technology in Chinese fencing teaching. In: *2022 2nd international conference on information technology and contemporary sports (TCS)*, pp.43–47. DOI: 10.1109/TCS56119.2022.9918801.
113. Liu J, Zheng Y, Wang K, et al. A real-time interactive tai chi learning system based on vr and motion capture technology. *Procedia Comput Sci* 2020; 174: 712–719.

114. Zhou S, Zhang F and Zheng Y. Research on wushu action recognition and matching based on kinect acquisition. *Techniques of Automation and Applications* 2020, in Chinese; 39: 94–97.
115. Cai X, Huo Y, Li F, et al. Human pose estimation and similarity calculation for tai chi learning. *J Graph* 2022; 43(04): 695–706.
116. Morel M, Achard C, Kulpa R, et al. Automatic evaluation of sports motion: a generic computation of spatial and temporal errors. *Image Vis Comput* 2017; 64: 67–78.
117. Xu Z. *Taiji boxing assist teaching and evaluation method based on whole body motion capture*. Master's Thesis, 2018, in Chinese.
118. Yang H and Li Y. Wushu movement evaluation method based on kinect. In: *2022 2nd international conference on information technology and contemporary sports (TCS)*, pp.29–33. DOI: 10.1109/TCS56119.2022.9918736.
119. Khan A, Haris M, Nadeem S, et al. Virtual self defense trainer -analyzing and scoring user pose. In: *2020 international congress on human-computer interaction, optimization and robotic applications (HORA)*, pp.1–5. DOI: 10.1109/HORA49412.2020.9152860.
120. Li H, Khoo S and Yap HJ. Differences in motion accuracy of baduanjin between novice and senior students on inertial sensor measurement systems. *Sensors* 2020; 20(21): 1–23. DOI: 10.3390/s20216258
121. Tang C, Chen X, Gong J, et al. Wmnn: wearables-based multi-column neural network for human activity recognition. *IEEE J Biomed Health Inform* 2023; 27(1): 339–350.
122. Pang Y, Wang Q and Zhang C. Time-frequency domain pattern analysis of tai chi 12 gong fa based on skeleton key points detection. In: *2021 international conference on neural networks, information and communication engineering*, vol. 11933, pp.431–436. SPIE.
123. Zhong M, Hung J, Yang Y, et al. Ga-svm classifying method applied to dynamic evaluation of taekwondo. In: *2016 international conference on advanced materials for science and engineering (ICAMSE)*, pp.534–537. DOI: 10.1109/ICAMSE.2016.7840191.
124. Wessa E, Ashraf A and Atia A. Can pose classification be used to teach kickboxing? In: *2021 international conference on electrical, computer and energy technologies (ICECET)*, pp.1–6. DOI: 10.1109/ICECET52533.2021.9698656.
125. Yamanaka S, Shima K, Mutoh A, et al. Evaluation of martial arts demonstration focusing on peak timing using acceleration data. In: *2021 IEEE 10th global conference on consumer electronics (GCCE)*, pp.337–341. DOI: 10.1109/GCCE53005.2021.9621988.
126. Yamanaka S, Shima K, Mutoh A, et al. Evaluation system for martial arts demonstration from smartphone sensor data using deep neural networks on noisy labels. In: *2023 IEEE international conference on consumer electronics (ICCE)*, pp.1–5. DOI: 10.1109/ICCE56470.2023.10043511.
127. Quintero Ovalle J, Stawarz K and Marzo Pérez A. Exploring the addition of audio input to wearable punch recognition. In: *Interacción'19: proceedings of the 2019 international conference on human computer interaction*, 2019, Donostia-San Sebastián, Spain. Association for Computing Machinery (ACM).
128. Zhang F and Zhou S. Intelligent recognition of athletes' wrong actions based on computer vision technology. *Adhes* 2020; 44(12): 82–85.
129. Emad B, Atef O, Shams Y, et al. Ikarate: Karate kata guidance system. *Procedia Comput Sci* 2020; 175: 149–156.
130. Emad B, Atef O, Shams Y, et al. Ikarate: Improving karate kata. *Procedia Comput Sci* 2020; 170: 466–473.
131. Hülsmann F, Göpfert JP, Hammer B, et al. Classification of motor errors to provide real-time feedback for sports coaching in virtual reality — a case study in squats and tai chi pushes. *Comput Graph* 2018; 76: 47–59. DOI: 10.1016/j.cag.2018.08.003
132. Jang WJ, Lee KK, Lee WJ, et al. Development of an inertial sensor module for categorizing anomalous kicks in taekwondo and monitoring the level of impact. *Sensors* 2022; 22(7): 2591.
133. Tuong Thanh N, Dang Tuyen N, Dung L, et al. Implementation of technical data analysis of skeleton extracted from camera kinect in grading movements of Vietnamese martial arts. In: *2017 international conference on advanced technologies for communications (ATC)*, pp.241–244. DOI: 10.1109/ATC.2017.8167625.
134. Li Z, Zhou L and Meng M. Research and development of an action evaluation intelligent system for eight-style taichi based on kinect 2.0. *Journal of Hangzhou Normal University (Natural Sciences Edition)* 2021; 20(04): 381–386.
135. Dai Z. *Recognition and evaluation of Taichi based on graph convolutional neural network*. Master's Thesis, 2022, in Chinese. DOI: 10.27005/d.cnki.gdzku.2022.002396.
136. Funk C, Nagendra S, Scott J, et al. Learning dynamics from kinematics: estimating 2d foot pressure maps from video frames, 2018.
137. Wiacek M, Tomasiuk R and Zubrzycki IZ. Correlations between anthropometric measurements and sports discipline aptitude. *Appl Sci* 2022; 12(12): 5932.
138. Le VH. Deep learning-based for human segmentation and tracking, 3d human pose estimation and action recognition on monocular video of mads dataset. *Multimed Tools Appl* 2023; 82(14): 20771–20818.
139. Soo J, Woods CT, Arjunan SP, et al. Identifying the performance characteristics explanatory of fight outcome in elite pencak silat matches. *Int J Perform Anal Sport* 2018; 18(6): 973–985.
140. Holmes B, McHale IG and Żychaluk K. Detecting individual preferences and erroneous verdicts in mixed martial arts judging using Bayesian hierarchical models. *Eur J Oper Res* 2024; 312(2): 733–745.
141. UFC stats, <http://ufcstats.com/statistics/events/completed> (accessed 30 March 2022).
142. Hachaj T, Ogiela M and Koptyra K. Learning from annotated video: an initial study based on oyama karate tournament recordings. In: *2015 10th international conference on P2P, parallel, grid, cloud and internet computing (3PGCIC)*, pp.358–362. IEEE.
143. Holmes B, McHale IG and Żychaluk K. A Markov chain model for forecasting results of mixed martial arts contests. *Int J Forecast* 2023; 39(2): 623–640.
144. Grzenda A, Speier W, Siddarth P, et al. Machine learning prediction of treatment outcome in late-life depression. *Front Psychiatry* 2021; 12: 1783.

145. Meng D, Guo H, Liang S, et al. Effectiveness of a hybrid exercise program on the physical abilities of frail elderly and explainable artificial-intelligence-based clinical assistance. *Int J Environ Res Public Health* 2022; 19(12): 6988.
146. Wu K, Li Y, Zou Y, et al. Tai chi increases functional connectivity and decreases chronic fatigue syndrome: a pilot intervention study with machine learning and fmri analysis. *PLoS One* 2022; 17(12): 1–20.
147. Guo H, Qiu J, Yi L, et al. Validation of accelerometers combined with machine learning for estimating energy expenditure of health qigong baduanjin: 620. *Med Sci Sports Exerc* 2021; 53(8S): 208–208.
148. Zhan X, Liu Y, Raymond S, et al. Rapid estimation of entire brain strain using deep learning models. *IEEE Trans Biomed Eng* 2021; 68(11): 3424–3434.
149. Falcone M, Yadav N, Poellabauer C, et al. Using isolated vowel sounds for classification of mild traumatic brain injury. In: *2013 IEEE international conference on acoustics, speech and signal processing*, pp.7577–7581. DOI: 10.1109/ICASSP.2013.6639136.
150. Zhan X, Liu Y, Cecchi NJ, et al. Finding the spatial covariation of brain deformation with principal component analysis. *IEEE Trans Biomed Eng* 2022; 69(10): 3205–3215.
151. Zhan X, Li Y, Liu Y, et al. Machine-learning-based head impact subtyping based on the spectral densities of the measurable head kinematics. *J Sport Health Sci* 2023; 12(5): 619–629.
152. Zhan X, Sun J, Liu Y, et al. Toward more accurate and generalizable brain deformation estimators for traumatic brain injury detection with unsupervised domain adaptation. *arXiv* 2023; 2306(02): 05255.
153. Starke S, Zhao Y, Zinno F, et al. Neural animation layering for synthesizing martial arts movements. *ACM Trans Graph* 2021; 40(4): 1–16.
154. Won J, Gopinath D and Hodgins J. Control strategies for physically simulated characters performing two-player competitive sports. *ACM Trans Graph* 2021; 40(4): 1–11.
155. Zhou Y, Li Z, Xiao S, et al. Auto-conditioned recurrent networks for extended complex human motion synthesis. *arXiv: Learning* 2017.
156. Li H, Dai J, Zeng R, et al. Foot-constrained spatial-temporal transformer for keyframe-based complex motion synthesis. *Comput Animat Virtual Worlds* 2024; 35(1): 1–13.
157. Sun R, Sun Y, Duan L, et al. Wushu competition field segmentation based on expectation-maximization algorithm. *China New Telecommunications* 2021; 23(01): 95–97.
158. Takata M, Nakamura Y, Torigoe Y, et al. Strikes-thrusts activity recognition using wrist sensor towards pervasive kendo support system. In: *2019 IEEE international conference on pervasive computing and communications workshops (PerCom Workshops)*, pp.243–248. DOI: 10.1109/PERCOMW.2019.8730861.
159. Pang Y. *Action recognition for plum blossom boxing (Meihua Quan) based on graph convolutional neural network*. Master's Thesis, 2023, in Chinese. DOI: 10.27340/d.cnki.gstxy.2023.000253.
160. Scott J, Collins R, Funk C, et al. 4D model-based spatiotemporal alignment of scripted taiji quan sequences. In: *2017 IEEE international conference on computer vision workshops (ICCVW)*, pp.795–804. DOI: 10.1109/ICCVW.2017.99.
161. Zeng Y, Cao M, Zhang H, et al. An automatic matching interactive and visualization system for virtual martial arts. In Xiao L and Xu D (eds.) *Thirteenth international conference on graphics and image processing (ICGIP 2021)*, p. 120832A, vol. 12083. International Society for Optics and Photonics, SPIE. DOI: 10.1117/12.2623402.
162. Malawski F and Kwolek B. Real-time action detection and analysis in fencing footwork. In: *2017 40th international conference on telecommunications and signal processing (TSP)*, pp.520–523. DOI: 10.1109/TSP.2017.8076041.
163. Hao Z, Duan Y, Dang X, et al. Human complex motion recognition method based on channel state information. *Comput Eng* 2020; 46: 286–293.
164. Hao Z, Dang J and Zhang D. A highly robust human motion recognition method based on channel state information. *Journal of Lanzhou Jiaotong University* 2020; 39: 37–44.
165. Li J, Hu H, Xing Q, et al. Tai chi action quality assessment and visual analysis with a consumer rgb-d camera. In: *2022 IEEE 24th international workshop on multimedia signal processing (MMSP)*, pp.1–6. DOI: 10.1109/MMSP55362.2022.9949464.
166. Mottaghi A, Soryani M and Seifi H. Action recognition in freestyle wrestling using silhouette-skeleton features. *Eng Sci Technol Int J* 2020; 23(4): 921–930.
167. Baumann J, Wessel R, Krüger B, et al. Action graph a versatile data structure for action recognition. In: *2014 international conference on computer graphics theory and applications (GRAPP)*, pp. 1–10.
168. Choi W, Li L, Sekiguchi H, et al. Recognition of gait motion by using data mining. In: *2013 13th international conference on control, automation and systems (ICCAS 2013)*, pp.1213–1216. DOI: 10.1109/ICCAS.2013.6704173.
169. Kasiri S, Fookes C, Sridharan S, et al. Fine-grained action recognition of boxing punches from depth imagery. *Comput Vis Image Underst* 2017; 159: 143–153.
170. de Goma J, Bustos M, Sebastian J, et al. Detection of taekwondo kicks using rgb-d sensors. In: *Proceedings of the 2019 3rd international conference on software and e-business*. ICSEB '19, pp.129–133, New York, NY: Association for Computing Machinery. DOI: 10.1145/3374549.3374576.
171. Zhu H and Pun C. Human action recognition with skeletal information from depth camera. In: *2013 IEEE international conference on information and automation (ICIA)*, pp.1082–1085. DOI: 10.1109/ICInfA.2013.6720456.
172. Torigoe Y, Nakamura Y, Fujimoto M, et al. Strike activity detection and recognition using inertial measurement unit towards kendo skill improvement support system. *Sens Mater* 2020; 32(2): 651.
173. Hachaj T, Piekarczyk M and Ogiela MR. Human actions analysis: templates generation, matching and visualization applied to motion capture of highly-skilled karate athletes. *Sensors* 2017; 17(11): 2590.
174. Vega-Martínez G, Mirabent-Amor D, Figueroa-Cavero F, et al. Experience of use of an artificial neural network as an auxiliary in the physical fitness classification of taekwondo athletes from Mexico city. In: *2022 global medical engineering physics exchanges/pan American*

- health care exchanges (GMEPE/PAHCE), pp.1–6. DOI: 10.1109/GMEPE/PAHCE55115.2022.9757793.
175. Khasanshin I. Application of an artificial neural network to automate the measurement of kinematic characteristics of punches in boxing. *Appl Sci* 2021; 11(3): 1223.
 176. Echeverria J and Santos OC. Toward modeling psychomotor performance in karate combats using computer vision pose estimation. *Sensors* 2021; 21(24): 8378.
 177. Nurahmadan I, Jayanta and Pradnyana I. Utilization of pose estimation and multilayer perceptron methods in the development of taekwondo martial arts independent learning. In: *2021 international conference on informatics, multimedia, cyber and information system (ICIMCIS)*, pp.267–272. DOI: 10.1109/ICIMCIS53775.2021.9699268.
 178. Sharma G and Uttam A. Multilayer neural network model for mixed martial arts winner prediction. In: *2021 5th international conference on information systems and computer networks (ISCON)*, pp.1–5. DOI: 10.1109/ISCON52037.2021.9702452.
 179. Grinsztajn L, Oyallon E and Varoquaux G. Why do tree-based models still outperform deep learning on typical tabular data? In: *36th conference on neural information processing systems (NeurIPS 2022) track on datasets and benchmarks. Advances in neural information processing*, New Orleans, United States.
 180. Mahdi F, Sandya Prasvita D and Theresiawati. Implementation of convolutional neural network algorithm for android-based classification of taekwondo martial arts. In: *2022 international conference on informatics, multimedia, cyber and information system (ICIMCIS)*, pp.407–412. DOI: 10.1109/ICIMCIS56303.2022.10017502.
 181. Hoang H, Ha C, Nguyen D, et al. Poses classification in a taekwondo lesson using skeleton data extracted from videos with shallow and deep learning architectures. In: *FDSE 2022: future data and security engineering. big data, security and privacy, smart city and industry 4.0 applications*, pp.447–461. Springer. DOI: 10.1007/978-981-19-8069-530.
 182. Yuan L, He Z, Wang Q, et al. Spatial transformer network with transfer learning for small-scale fine-grained skeleton-based tai chi action recognition, pp.1–6. DOI: 10.1109/IECON49645.2022.9968668.
 183. Sato K, Matsubara H and Suzuki K. Commonality of motions having effective features with respect to methods for identifying the moves made during kumite sparring in karate. In: *8th international conference on sport sciences research and technology support (icSPORTS 2020)*, pp.75–82. DOI: 10.5220/0010132400750082.
 184. Jiang K. *Research on wushu action recognition based on deep learning*. Master's Thesis, 2019, in Chinese. DOI: 10.27312/d.cnki.gshsu.2022.000531.
 185. Ohishi Y, Tanaka Y and Kashino K. Unsupervised co-segmentation for athlete movements and live commentaries using crossmodal temporal proximity. In: *2020 25th international conference on pattern recognition (ICPR)*, pp.9137–9142. DOI: 10.1109/ICPR48806.2021.9412233.
 186. Labintsev A, Khasanshin I, Balashov D, et al. Recognition punches in karate using acceleration sensors and convolution neural networks. *IEEE Access* 2021; 9: 138106–138119.
 187. Chen Y. *Research on action analysis based on human skeleton information*. Master's Thesis, 2021, in Chinese. DOI: 10.27075/d.cnki.ghzdc.2021.000196.
 188. Ruan W, Zhang B, Lu S, et al. Stance recognition applied to Chinese martial arts long stick combat sports. In: *2023 IEEE 3rd international conference on electronic technology, communication and information (ICETCI)*, pp.1780–1783. DOI: 10.1109/ICETCI57876.2023. 10176867.
 189. Liang J and Zuo G. Taekwondo action recognition method based on partial perception structure graph convolution framework. *Sci Program* 2022; 2022: 1–10.
 190. Tung PS, Giang PH, et al.; Do TT. Support learning vovinam exercises based on computer vision. In: Rutkowski L, Scherer R and Korytkowski M (eds) *Artificial intelligence and soft computing*. Cham: Springer Nature Switzerland, 2023, pp.102–111.
 191. Tejero-de Pablos A, Nakashima Y, Sato T, et al. Summarization of user-generated sports video by using deep action recognition features. *IEEE Trans Multimedia* 2018; 20(8): 2000–2011.
 192. Tejero-de Pablos A, Nakashima Y, Sato T, et al. Human action recognition-based video summarization for rgb-d personal sports video. In: *2016 IEEE international conference on multimedia and expo (ICME)*, pp.1–6. DOI: 10.1109/ICME.2016.7552938.
 193. Hou Y, Seydou FM and Kenderdine S. Unlocking a multimodal archive of southern Chinese martial arts through embodied cues. *J Doc* 2023; 80: 1148–1166. DOI: 10.1108/JD-01-2022-0027
 194. Min Y. *Research on sports auxiliary training system based on multi-kinect*. Master's Thesis, 2021, in Chinese. DOI: 10.27390/d.cnki.gxbfc.2021.000064.
 195. Min Y and Mou L. Research on sports auxiliary training system based on multiple kinect. *Electron Des Eng* 2021; 29(18): 115–120.
 196. Behendi S, Morgan S and Fookes C. Non-invasive performance measurement in combat sports. In Chung P, Soltoggio A, Dawson CW, et al. (eds.) *Proceedings of the 10th international symposium on computer science in sports (ISCSS)*, pp.3–10. Cham: Springer International Publishing. ISBN 978-3-319-24560-7.
 197. Kasiri-Bidhendi S, Fookes C, Morgan S, et al. Combat sports analytics: Boxing punch classification using overhead depthimager. In: *2015 IEEE international conference on image processing (ICIP)*, pp.4545–4549. DOI: 10.1109/ICIP.2015.7351667.
 198. Zhu K, Wong A and McPhee J. Fencenet: fine-grained footwork recognition in fencing. In: *2022 IEEE/CVF conference on computer vision and pattern recognition workshops (CVPRW)*, pp.3588–3597. DOI: 10.1109/CVPRW56347.2022.00403.
 199. Barbosa P, Cunha P, Carvalho V, et al. Deep learning in taekwondo techniques recognition system: a preliminary approach. In: Machado J, Soares F and Trojanowska J (eds) *Innovations in mechatronics engineering II*. Cham: Springer International Publishing, 2022, pp. 280–291.
 200. Cunha P, Barbosa P, Ferreira F, et al. Real-time evaluation system for top taekwondo athletes: project overview. In: *International conference on biomedical electronics and devices*. DOI: 10.5220/0010414200002865.
 201. Barbosa P, Cunha P, Carvalho V, et al. Classification of taekwondo techniques using deep learning methods:

- first insights. In: *Proceedings of the 14th international joint conference on biomedical engineering systems and technologies (BIOSTEC 2021) -BIODEVICES*. INSTICC, pp.201–208. SciTePress. DOI: 10.5220/0010412400002865.
202. Nakamura T, Nagai T, Mochihashi D, et al. Segmenting continuous motions with hidden semi-Markov models and gaussian processes. *Front Neurorobot* 2017; 11: 67.
 203. Daker M, Elsayaad F and Atia A. Karate kata style classification using pose landmarks and deep learning. In: *2023 5th novel intelligent and leading emerging sciences conference (NILES)*, pp.228–231. DOI: 10.1109/NILES59815.2023.10296806.
 204. Dong J, Fang Q, Jiang W, et al. Fast and robust multi-person 3d pose estimation and tracking from multiple views. *IEEE Trans Pattern Anal Mach Intell* 2022; 44(10): 6981–6992.
 205. Peng S, Zhang Y, Xu Y, et al. Neural body: implicit neural representations with structured latent codes for novel view synthesis of dynamic humans. In: *2021 IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pp.9050–9059. DOI: 10.1109/CVPR46437.2021.00894.
 206. Li Z, Liu J, Zhang Z, et al. Cliff: carrying location information in full frames into human pose and shape estimation. In: Avidan S, Brostow G and Cisse M (eds) *Computer Vision – ECCV 2022*. Cham: Springer Nature Switzerland, 2022, pp. 590–606.
 207. Hong J, Fisher M, Gharbi M, et al. Video pose distillation for few-shot, fine-grained sports action recognition. In: *2021 IEEE/CVF international conference on computer vision (ICCV)*, pp.9234–9243. DOI: 10.1109/ICCV48922.2021.00912.
 208. Hirose S, Kato T, Yamashita T, et al. Action Quality Assessment Model using specialists' gaze location and kinematics data—focusing on evaluating figure skating jumps. *Sensors* 2023; 23(22): 9282.
 209. Kim J, Kim H, Lee J, et al. A deep learning approach for fatigue prediction in sports using GPS data and rate of perceived exertion. *IEEE Access* 2022; 10: 103056–103064.
 210. Halilaj E, Rajagopal A, Fiterau M, et al. Machine learning in human movement biomechanics: best practices, common pitfalls, and new opportunities. *J Biomech* 2018; 81: 1–11.
 211. Rommers N, Rössler R, Verhagen E, et al. A machine learning approach to assess injury risk in elite youth football players. *Med Sci Sports Exerc* 2020; 52(8): 1745–1751.
 212. Wang Y, Liu W and Liu X. Explainable ai techniques with application to NBA gameplay prediction. *Neurocomputing* 2022; 483: 59–71.